

Template Week 5 – Operating Systems

Student number: 583809

Assignment 5.1: Unix-like

- a) Find out what the difference is between UNIX and unix-like operating systems?
UNIX is gecertificeerd officieel besturingsstelsel, daarbij zijn die unixe like dat niet. Unix like zijn vaak opensource gebaseerd op unix concepten.
- b) Study the image above named UNIX timeline. Find out who Ken Thompson, Dennis Ritchie, Bill Joy, Richard Stallman, and Linus Torvalds are and what they have contributed to the development of UNIX or unix-like systems and to IT in general. **TIP!** English-language sources often contain more detailed information about these individuals.

Ken Thompson is een mede uitvinder van UNIX. Hij schreef onder anderen de eerste unix kernel en ontwikkelde de programmeertaal B (Voorloper van C)

Dennis Ritchie is ook een mede oprichter van unix samen met Ken. Hij ontwikkelde de programmeertaal C waarop alle moderne systemen zijn gebaseerd.

Bill Joy is medeoprichter van Sun Microsystems, hoofd ontwikkelaar van BSD Unix en maker van vi-editor wat de voorloper van vim was.

Richard Stallman was oprichter van GNU beweging, die ook startte de Free Software Foundation en wilde een volledig vrije Unix achtige systeem bouwen. Schreef onder anderen ook belangrijke tool zoals gcc, emacs en de GNU Licentie (GPL)

Linus Torvalds is de maker van Linux kernel, wat combineerde zijn kernel met GNU software, GNU/Linux

Linux is nu op veel besturingsystemen de basis zoals Android en servers

- c) What is the philosophy of the GNU movement?
Dat het vrij is, maar niet perse gratis.
Vrijheid om software te gebruiken
Vrijheid om software te besturen
Vrijheid om software te wijzigen
Vrijheid om software te verspreiden met of zonder aanpassingen
- d) Does Ubuntu as a Linux operating system conform to the philosophy of the GNU movement?
Please explain your answer.
Deels maar niet volledig.
Het is gebaseerd op Linux/GNU
Het is ook open source en heeft ook de GPL licenties
- Ubuntu wijkt wel af van de filosofie
Ze maken soms commerciële keuzes
Ze bieden soms proprietary software aan in de repositories

- e) Find out what is the Windows Subsystem for Linux?

Wsl is een functie in windows waarmee je linux programma's kunt draaien binnen windows zonder virtuele machine of dual-boot. Dit is handig voor vele developers, scripting en serversimulatie.

- f) Find out, which operating system family belongs to Android, iOS and ChromeOS?

Android = linux familie, gebaseerd op de linux kernel

ios = Unix familie gebaseerd op darwin, afgeleid van bsd unix

ChromeOS= linux familie, gebaseerd op gentoo linux.

Assignment 5.2: Supercomputers and gameconsoles

- a) Research on this site what supercomputers are used for and write a short summary of it:

De eerste supercomputer wat gebruikt maakt van linux os, wat gemaakt door David A Bader terwijl hij op de universiteit of New Mexico. Dit was een prototype van de linux gebaseerde supercomputer. De prototype kwam in productie in april 1999.

In 1997 hebben onderzoekers een kampioens schaak programma gemaakt en die tussen de kampioen laten spelen. Die schaakprogramma heet IBM's Deep Blue chess supercomputer, hij heeft de kampioen speler meerdere keren verslagen in het spel. Twee keer verloren de kampioen, drie keer gelijkspel, waarbij er word gezegd dat mensen ermee hebben bemoeit. IDB heeft de supercomputer in pensioen gezet.

<https://www.computerhistory.org/timeline/1997/#169ebbe2ad45559efbc6eb357207f4d9>
<https://www.computerhistory.org/timeline/1998/#16f0dbd15d916dbc4ac105c55c000936>
<https://www.computerhistory.org/timeline/search/?q=Supercomputer>

- b) IBM is a company that has already built a number of supercomputers. One of them is IBM's Roadrunner. The CPU developed for this supercomputer was further developed at a later stage as the CPU for the PlayStation 3 console. Find out what a **PlayStation 3 cluster** is and what it was used for?

De playstation 3 cluster is een distributie system (verspreiden), om te zorgen dat heel veel ps3 consoles via de netwerk te koppelen. Ze deden dit omdat het een goedkope manier was om veel rekenkracht te hebben. Het amerikaanse leger, universiteiten, simulaties, etc. dit hebben ze vaak met linux gedaan op de PS3.

- c) You can build a supercomputer by putting a few computers together in a cluster. Here's what Oracle did with a collection of Raspberry Pi's, for example:

<https://blogs.oracle.com/developers/post/building-the-worlds-largest-raspberry-pi-cluster>

What specific operating system is running on this cluster?
Oracle linux for ARM (aarch64).

- d) Does Oracle's Raspberry Pi supercomputer appear in the list of the 500 fastest supercomputers in the world? Make a logical decision for this, without going through the entire list.

<https://www.top500.org/lists/top500/list/2023/06/>

Ik denk van niet, omdat de Raspberry Pi CPU zijn bedoelt voor lichte taken/ educatie, niet voor extreme hpc performance (high performance computing).

- e) What CPU architecture is used for the PlayStation 5 and Xbox Series X?

Hun hebben beide AMD zen 2 8-core, x86-64

What operating systems run on these consoles?

Playstation 5 = orbis OS (wat gebaseerd is op FreeBSD, een Unix like os) xbox series x = Microsoft Windows os (is gebaseerd daarop en bevat ook nog DirectX-functies die beter zijn gemaakt voor consoles)

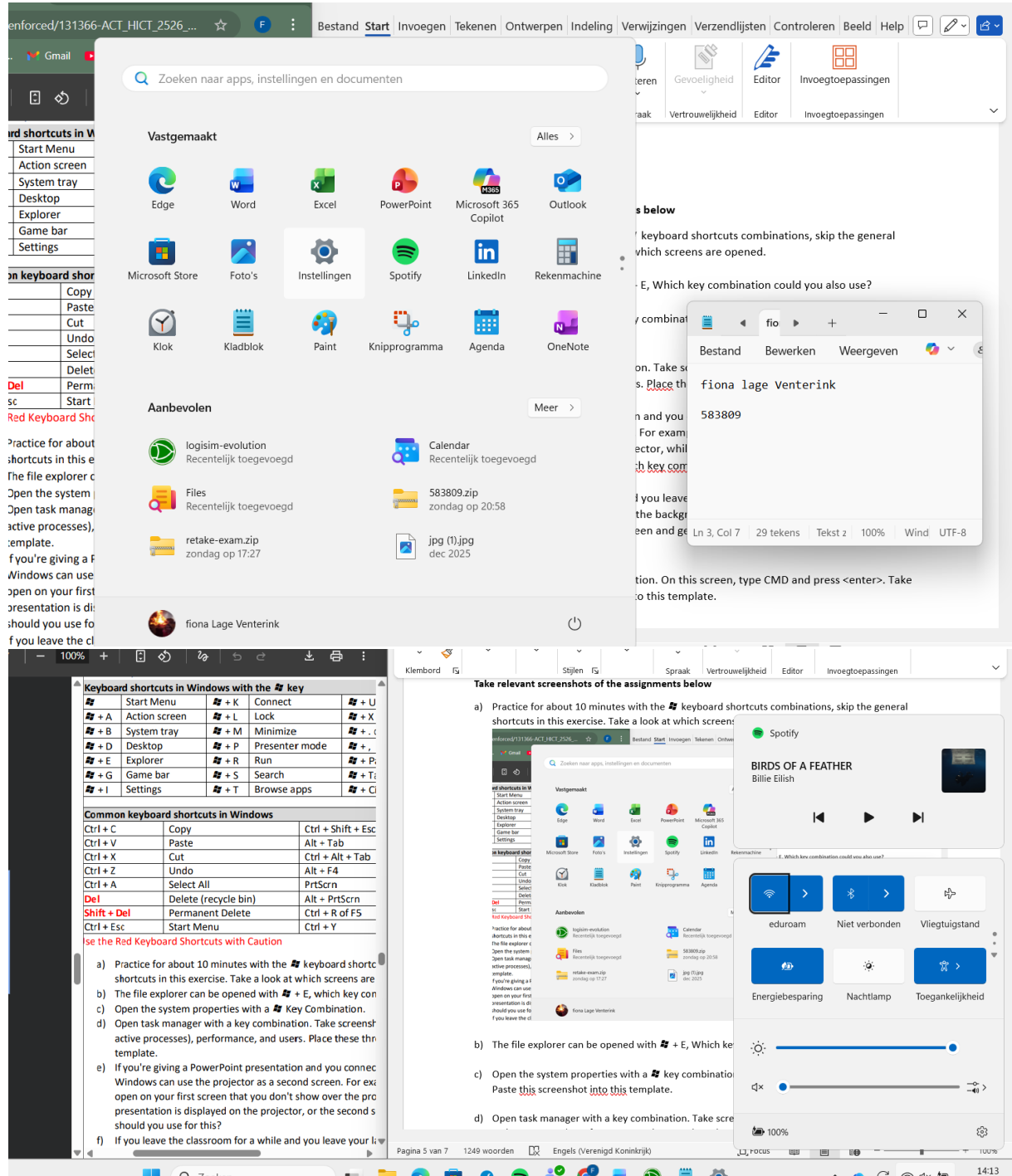
What conclusion can you draw from the answer to the previous question?

De consoles gebruiken pc achtige hardware en besturingsystemen die erg lijken op de besturingsystemen. Dit doen ze groot waarschijnlijk bewust om de producten op meerdere platvorms te hebben.

Assignment 5.3: Working with Windows

Take relevant screenshots of the assignments below

- a) Practice for about 10 minutes with the  keyboard shortcuts combinations, skip the general shortcuts in this exercise. Take a look at which screens are opened.



The screenshot displays a Windows 11 desktop environment. On the left, a sidebar lists various keyboard shortcuts categorized by function (e.g., Start Menu, Action screen, System tray, Desktop, Explorer, Game bar, Settings). The main area shows the Start menu with a search bar and a grid of app tiles including Edge, Word, Excel, PowerPoint, Microsoft 365 Copilot, Outlook, Microsoft Store, Foto's, Instellingen, Spotify, LinkedIn, Rekenmachine, Klok, Kladblok, Paint, Knipprogramma, Agenda, and OneNote. Below the Start menu, a list of recommended apps and files is visible, including 'logisim-evolution', 'Calendar', 'Files', '583809.zip', 'retake-exam.zip', and 'jpg (1).jpg'. A small window titled 'fio' is open in the foreground, showing a text editor with the content 'fiona lage Venterink' and '583809'. The taskbar at the bottom shows the Start button, a search bar, and several pinned apps. The system tray on the right includes icons for network, volume, and battery status.

Keyboard shortcuts in Windows with the  key

Shortcut	Action	Shortcut	Action
 + S	Start Menu	 + K	Connect
 + A	Action screen	 + L	Lock
 + B	System tray	 + M	Minimize
 + D	Desktop	 + P	Presenter mode
 + E	Explorer	 + R	Run
 + G	Game bar	 + S	Search
 + I	Settings	 + T	Browse apps

Common keyboard shortcuts in Windows

Shortcut	Action	Shortcut	Action
Ctrl + C	Copy	Ctrl + Shift + Esc	Task Manager
Ctrl + V	Paste	Alt + Tab	Switch between apps
Ctrl + X	Cut	Ctrl + Alt + Tab	Switch between apps
Ctrl + Z	Undo	Alt + F4	Close current app
Ctrl + A	Select All	PrtScr	Print screen
Del	Delete (recycle bin)	Alt + PrtScr	Print screen and copy to clipboard
Shift + Del	Permanent Delete	Ctrl + R or F5	Refresh
Ctrl + Esc	Start Menu	Ctrl + Y	Redo

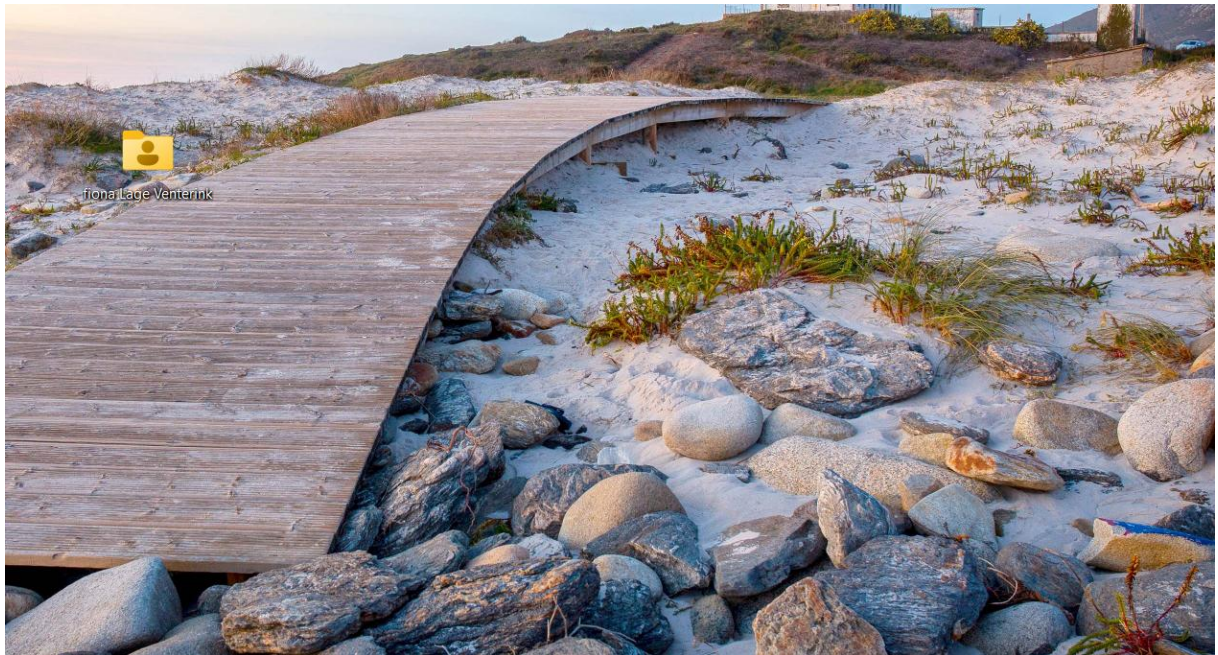
Take relevant screenshots of the assignments below

a) Practice for about 10 minutes with the  keyboard shortcuts combinations, skip the general shortcuts in this exercise. Take a look at which screens are opened.

b) The file explorer can be opened with  + E, Which key combination could you also use?

c) Open the system properties with a  key combination. Paste this screenshot into this template.

d) Open task manager with a key combination. Take screenshot of this screen.



- b) The file explorer can be opened with **Windows + E**, Which key combination could you also use?
Windows + R -> typ explorer -> enter
- c) Open the system properties with a **Windows** key combination, take a screenshot of the open screen.
Paste this screenshot into this template.

with the **Windows** key

Connect	Windows + U
Lock	Windows + X
Minimize	Windows + . of ;
Presenter mode	Windows + ,
Run	Windows + Paus
Search	Windows + Tab
Browse apps	Windows + Cijfer

Windows

Ctrl + Shift + Esc
Alt + Tab
Ctrl + Alt + Tab
Alt + F4
PrtScr
Alt + PrtScr
Ctrl + R of F5
Ctrl + Y

Caution

a) Practice for about 10 minutes with the **Windows** keyboard shortcuts in this exercise. Take a look at which screens are open.

b) The file explorer can be opened with **Windows** + E, which key combination could you also use?

c) Open the system properties with a **Windows** key combination.

d) Open task manager with a key combination. Take screenshots of the tabs: processes (shows active processes), performance, and users. Place these three screenshots in this template.

e) If you're giving a PowerPoint presentation and you connect your laptop to a projector, Windows can use the projector as a second screen. For example, you may have Outlook open on your first screen that you don't show over the projector, while the PowerPoint presentation is displayed on the projector, or the second screen. Which key combination should you use for this?

f) If you leave the classroom for a while and you leave your laptop behind, it is wise to lock the screen. Your Apps will continue to run in the background. So, for example, if you're waiting for a download that takes a while, lock the screen and get a cup of coffee. Which key combination do you use for this?

- d) Open task manager with a key combination. Take screenshots of the tabs: processes (shows active processes), performance, and users. Place these three screenshots in this template.

Taakbeheer

Processen Nieuwe taak uitvoeren Taak beëindigen Efficiëntiemodus

Naam	Status	43% Processor	86% Geheugen	2% Schijf	0% Netwerk
Toepassingen (6)					
Google Chrome (13)		13.2%	991.1 MB	0 MB/s	0 Mbps
Instellingen		0%	0 MB	0 MB/s	0 Mbps
Microsoft Teams		0%	21.4 MB	0 MB/s	0 Mbps
Notepad.exe		0%	7.9 MB	0 MB/s	0 Mbps
Taakbeheer		6.0%	41.4 MB	0.8 MB/s	0 Mbps
WINWORD.EXE		0%	109.8 MB	0 MB/s	0 Mbps
Achtergrondprocessen (69)					
AggregatorHost.exe		0%	1.4 MB	0 MB/s	0 Mbps
App-acties		0%	2.3 MB	0 MB/s	0 Mbps
audiodg.exe		0%	4.6 MB	0 MB/s	0 Mbps
Background Task Host		0%	0.8 MB	0 MB/s	0 Mbps
Crashpad		0%	0.4 MB	0 MB/s	0 Mbps
Credential Guard & VBS Key Is...		0%	0.3 MB	0 MB/s	0 Mbps
Dell Data Vault Data Collector...		0%	0.4 MB	0 MB/s	0 Mbps
Dell Data Vault Rules Processor		0%	0.8 MB	0 MB/s	0 Mbps
Dell SupportAssist Remediation		0%	4.9 MB	0 MB/s	0 Mbps
Digital logic design tool and s...		0%	0.3 MB	0 MB/s	0 Mbps
Discord		0%	46.4 MB	0 MB/s	0 Mbps
Discord		0%	4.6 MB	0 MB/s	0 Mbps
Discord.exe		0%	1.3 MB	0 MB/s	0 Mbps
DSAPL.exe		0%	7.9 MB	0 MB/s	0 Mbps
EA		0%	1.1 MB	0 MB/s	0 Mbps

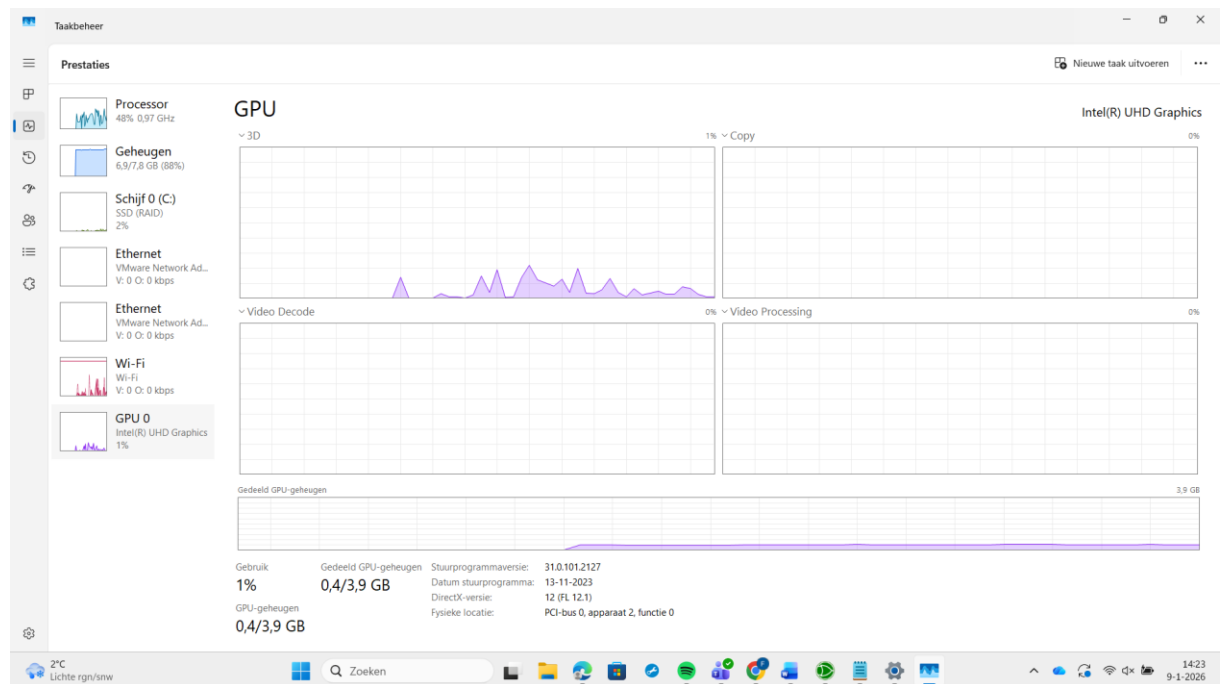
2°C Lichte rgn/snw 14:22 9-1-2026

Taakbeheer

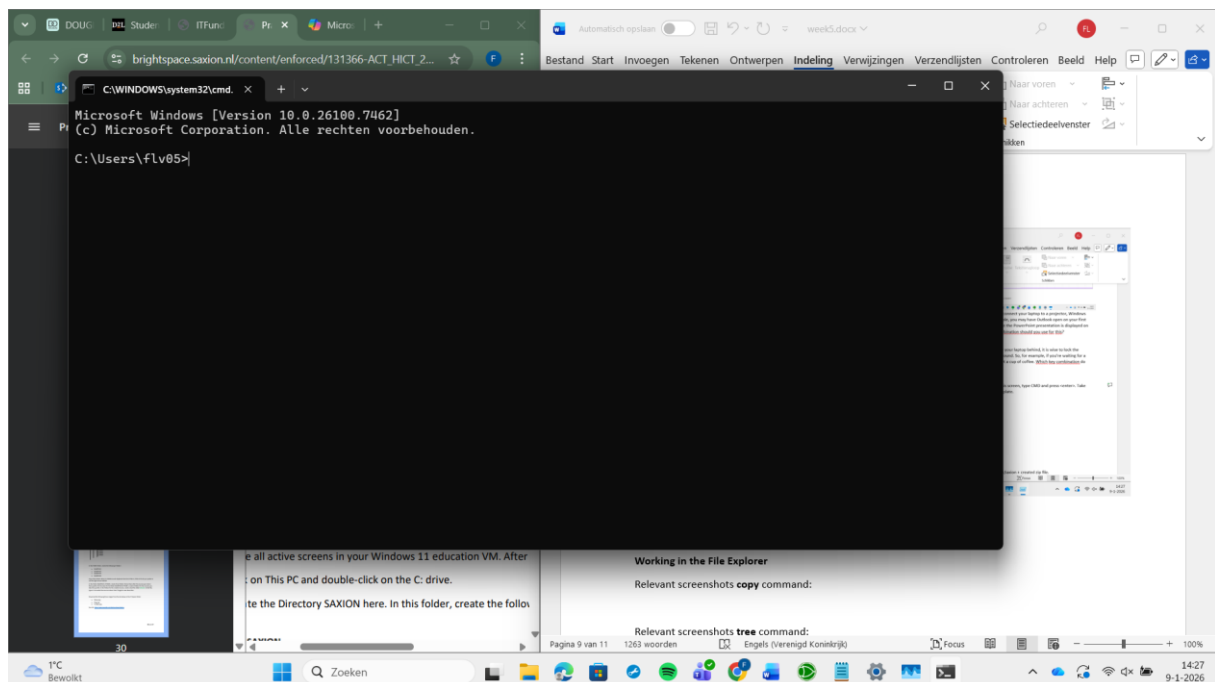
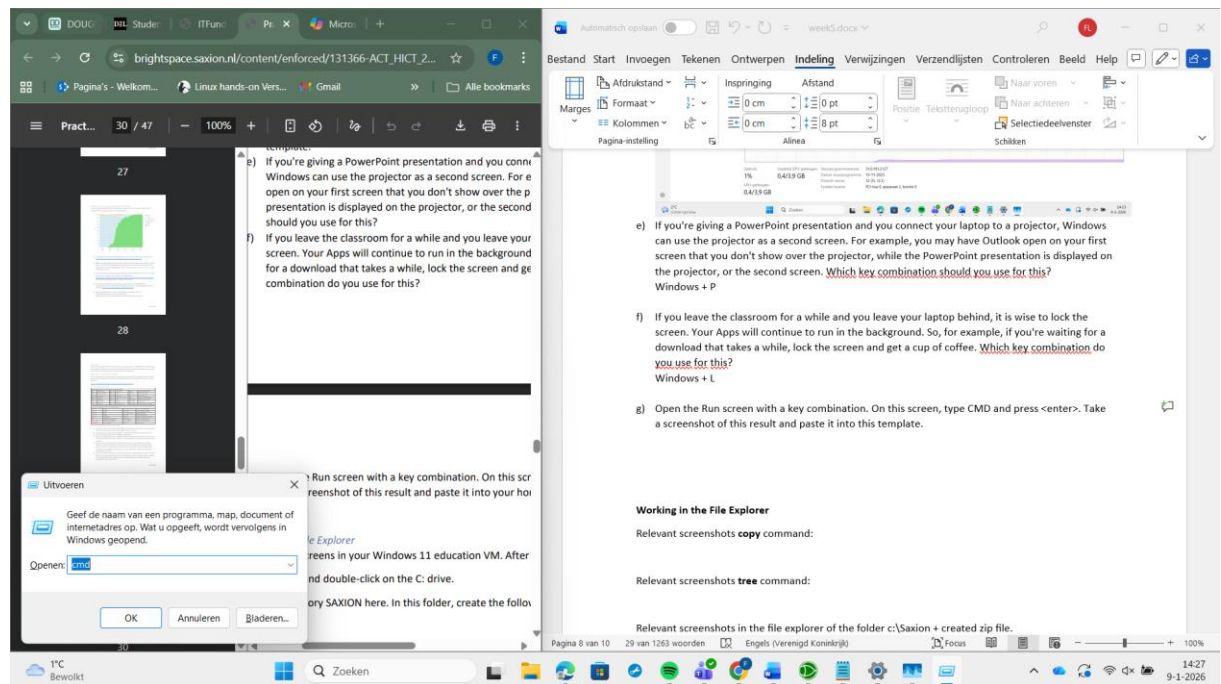
Gebruikers Nieuwe taak uitvoeren Verbinding verbreken Gebruikersaccounts beheren

Gebruiker	Status	46% Processor	87% Geheugen	2% Schijf	0% Netwerk
fiv05@hotmail.com (148)		25.9%	2,244.4 MB	0.1 MB/s	0.1 Mbps

2°C Lichte rgn/snw 14:23 9-1-2026

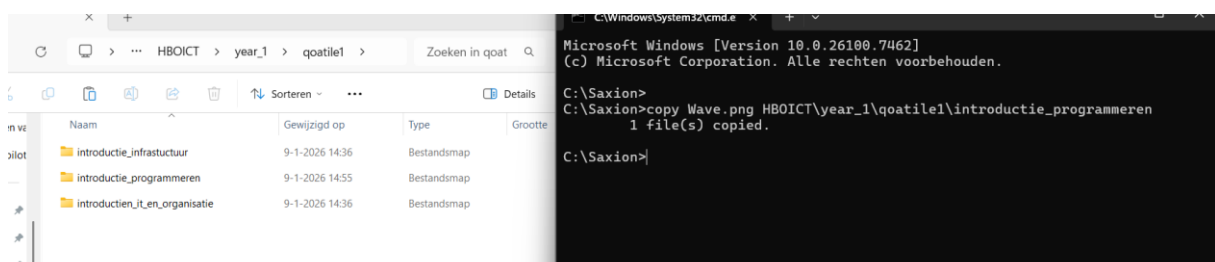


- e) If you're giving a PowerPoint presentation and you connect your laptop to a projector, Windows can use the projector as a second screen. For example, you may have Outlook open on your first screen that you don't show over the projector, while the PowerPoint presentation is displayed on the projector, or the second screen. Which key combination should you use for this?
Windows + P
- f) If you leave the classroom for a while and you leave your laptop behind, it is wise to lock the screen. Your Apps will continue to run in the background. So, for example, if you're waiting for a download that takes a while, lock the screen and get a cup of coffee. Which key combination do you use for this?
Windows + L
- g) Open the Run screen with a key combination. On this screen, type CMD and press <enter>. Take a screenshot of this result and paste it into this template.



Working in the File Explorer

Relevant screenshots **copy** command:



```
1 file(s) copied.

C:\Saxion>copy Plug.png HBOICT\year_1\qoatile1\introduction_infrastructures
1 file(s) copied.

C:\Saxion>

C:\Saxion>copy Tumble.png HBOICT\year_1\qoatile1\introduction_it_en_organisa
tie
1 file(s) copied.

C:\Saxion>
```

Relevant screenshots **tree** command:



20260109-1402-56.38
90363.mp4

C:.

```
├──HBOICT
|   ├──year_1
|   |   ├──qoatile1
|   |   |   ├──introduction_it_en_organisatie
|   |   |   |   ├──locales
|   |   |   |   ├──resources
|   |   |   |   |   ├──app.asar.unpacked
|   |   |   |   |   |   ├──node_modules
|   |   |   |   |   |   ├──vscode-windows-ca-certs
|   |   |   |   |   |   |   ├──build
|   |   |   |   |   |   |   ├──Release
|   |   |   |   |   |   |   |   ├──obj
|   |   |   |   |   |   |   |   ├──crypt32
|   |   |   |   |   |   |   |   ├──crypt32.tlog
|   |   |   |   |   |   |   |   ├──win-ca
|   |   |   |   |   |   |   |   ├──api
|   |   |   |   |   |   |   |   ├──fallback
|   |   |   |   |   |   |   |   ├──lib
```

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| | | | └─support
| | | └─introdactie_infrastuctuur
| | | └─introdactie_programmeren
| | | └─.idea
| | | | └─libraries
| | | └─251014_ACT-IP01-frm
| | | | └─.idea
| | | | | └─libraries
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						└─Exercise5
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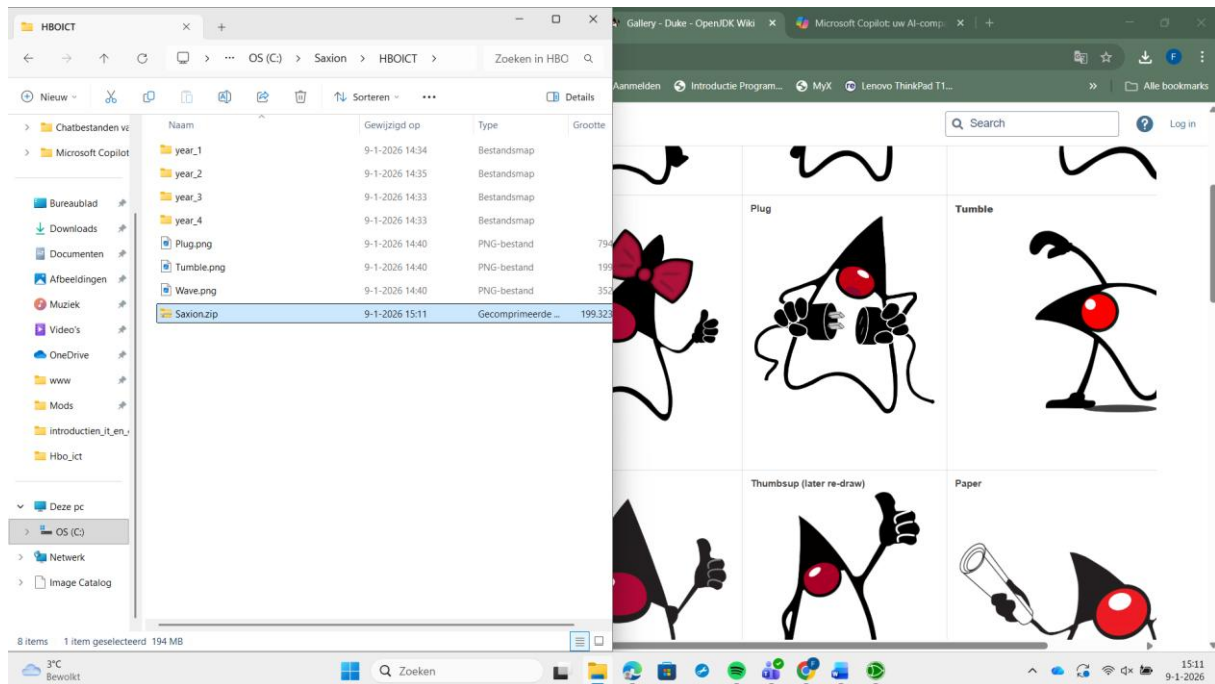
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[illegible]

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					└─Exercise14
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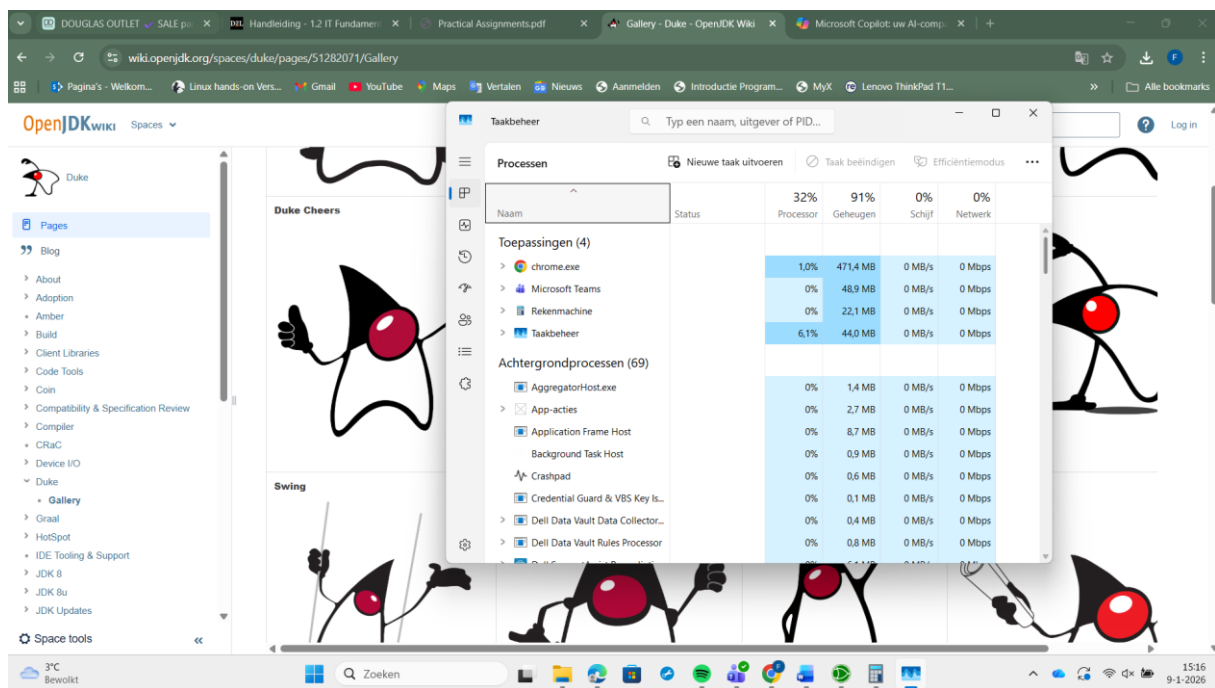
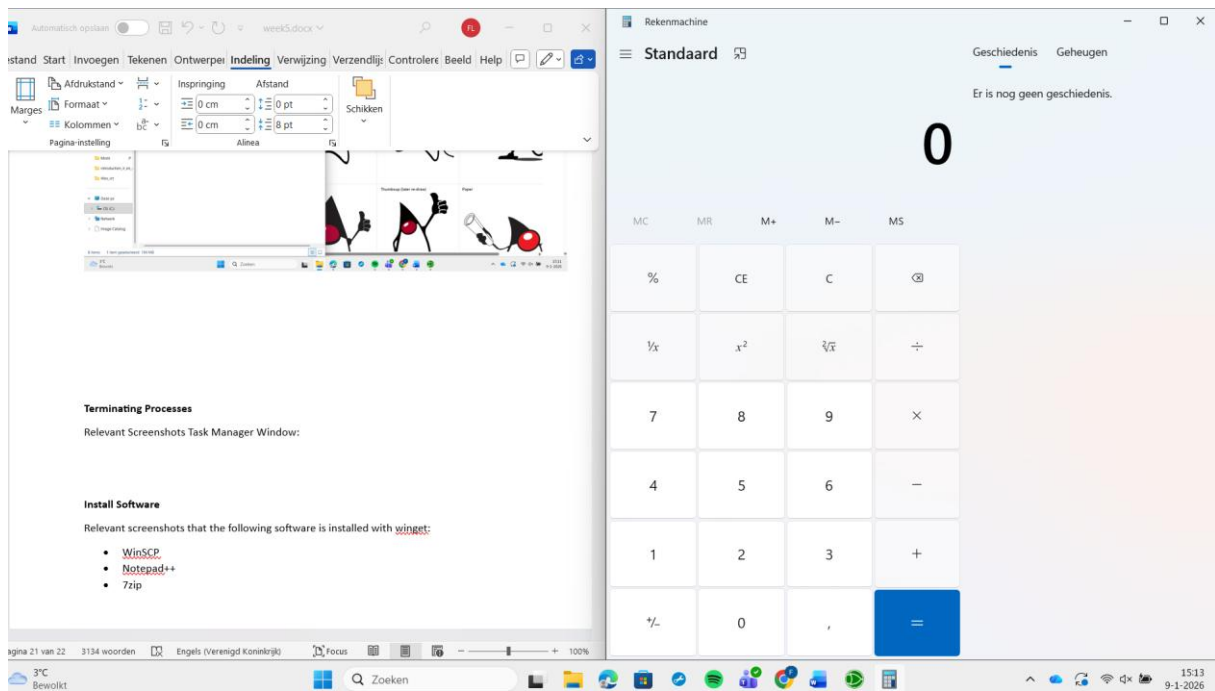
```
C:\Saxion>echo %username%
flv05
```

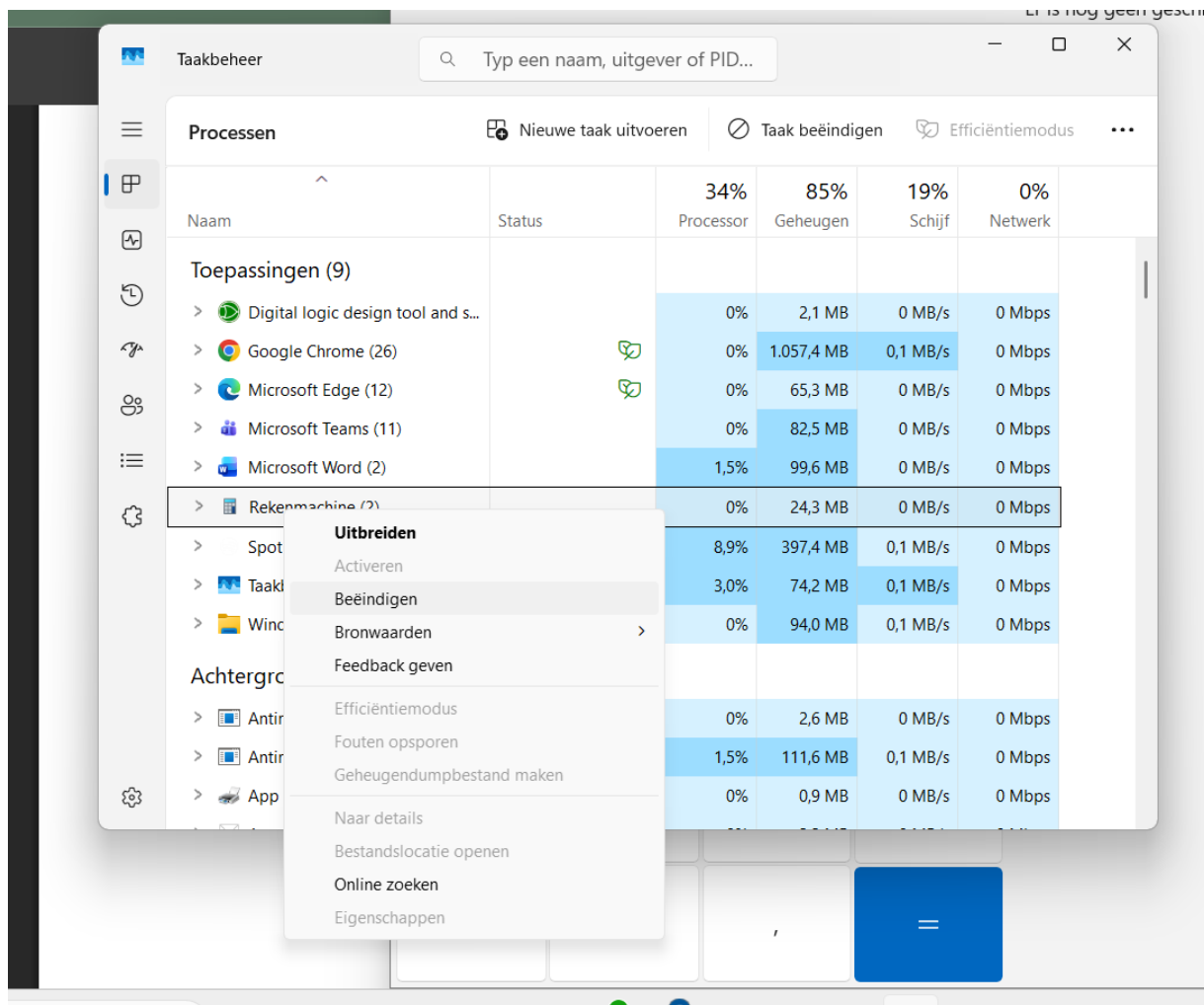
Relevant screenshots in the file explorer of the folder c:\Saxion + created zip file.



Terminating Processes

Relevant Screenshots Task Manager Window:





Install Software

Relevant screenshots that the following software is installed with winget:

Ik heb gewoon bij firefox install gedaan want dat was makkelijker


```
Administrator: Opdrachtprompt
Microsoft Windows [Version 10.0.26100.7462]
(c) Microsoft Corporation. Alle rechten voorbehouden.

C:\Windows\System32>winget
Windows Package Manager v1.12.350
Copyright (c) Microsoft Corporation. All rights reserved.

The winget command line utility enables installing applications and other packages from the command line.

usage: winget [<command>] [<options>]

The following commands are available:
install    Installs the given package
show       Shows information about a package
source     Manage sources of packages
search     Find and show basic info of packages
list       Display installed packages
upgrade    Shows and performs available upgrades
uninstall  Uninstalls the given package
hash       Helper to hash installer files
validate   Validates a manifest file
settings   Open settings or set administrator settings
features   Shows the status of experimental features
export     Exports a list of the installed packages
import     Installs all the packages in a file
pin        Manage package pins
configure  Configures the system into a desired state
download   Downloads the installer from a given package
repair     Repairs the selected package
dscv3      DSC v3 resource commands
mcp        MCP information

For more details on a specific command, pass it the help argument. [-?]

The following options are available:
-v, --version      Display the version of the tool
--info             Display general info of the tool
-?, --help         Shows help about the selected command
--wait            Prompts the user to press any key before exiting
--logs, --open-logs Open the default logs location
--verbose, --verbose-logs Enables verbose logging for winget
--nowarn, --ignore-warnings Suppresses warning outputs
--disable-interactivity Disable interactive prompts
--proxy           Set a proxy to use for this execution
--no-proxy        Disable the use of proxy for this execution

More help can be found at: https://aka.ms/winget-command-help

C:\Windows\System32>
```

```
More help can be found at: https://aka.ms/winget-command-help

C:\Windows\System32>winget install Mozilla.Firefox
The 'msstore' source requires that you view the following agreements before using.
Terms of Transaction: https://aka.ms/microsoft-store-terms-of-transaction
The source requires the current machine's 2-letter geographic region to be sent to the backend service to function properly (ex. "US").

Do you agree to all the source agreements terms?
[Y] Yes [N] No:
[Y] Yes [N] No: y
Found Mozilla Firefox (en-US) [Mozilla.Firefox] Version 146.0.1
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://download-installer.cdn.mozilla.net/pub/firefox/releases/146.0.1/win64/en-US/Firefox%20Setup%20146.0.1.exe
[Progress Bar] 82.3 MB / 82.3 MB
Successfully verified installer hash
Starting package install...
Successfully installed

C:\Windows\System32>
```

```
C:\Windows\System32>winget install WinSCP.WinSCP
Found WinSCP [WinSCP.WinSCP] Version 6.5.5
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://sourceforge.net/projects/winscp/files/WinSCP/6.5.5/WinSCP-6.5.5-Setup.exe/download
[Progress Bar] 11.6 MB / 11.6 MB
Successfully verified installer hash
Starting package install...
Successfully installed

C:\Windows\System32>
C:\Windows\System32>
```

```
C:\Windows\System32>winget install Notepad++.Notepad++  
Found Notepad++ [Notepad++.Notepad++] Version 8.9  
This application is licensed to you by its owner.  
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.  
Downloading https://github.com/notepad-plus-plus/notepad-plus-plus/releases/download/v8.9/npp.8.9.Installer.x64.exe  
██████████████████████████████████████████████████████ 6.54 MB / 6.54 MB  
Successfully verified installer hash  
Starting package install...  
Successfully installed
```

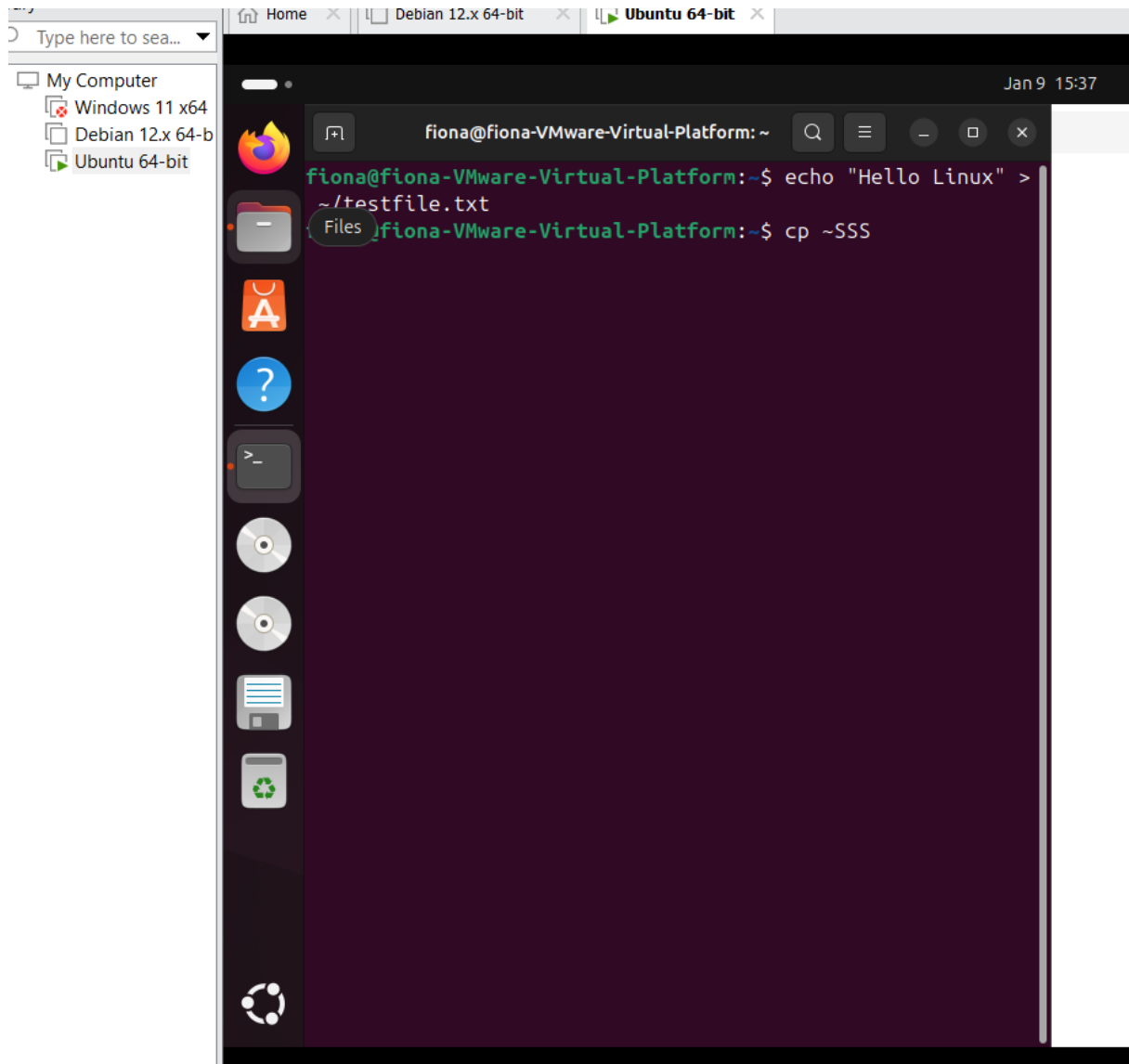
```
C:\Windows\System32>  
C:\Windows\System32>  
C:\Windows\System32>winget install 7zip.7zip  
Found 7-Zip [7zip.7zip] Version 25.01  
This application is licensed to you by its owner.  
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.  
Downloading https://7-zip.org/a/7z2501-x64.exe  
██████████████████████████████████████████████████████ 1.56 MB / 1.56 MB  
Successfully verified installer hash  
Starting package install...  
Successfully installed
```

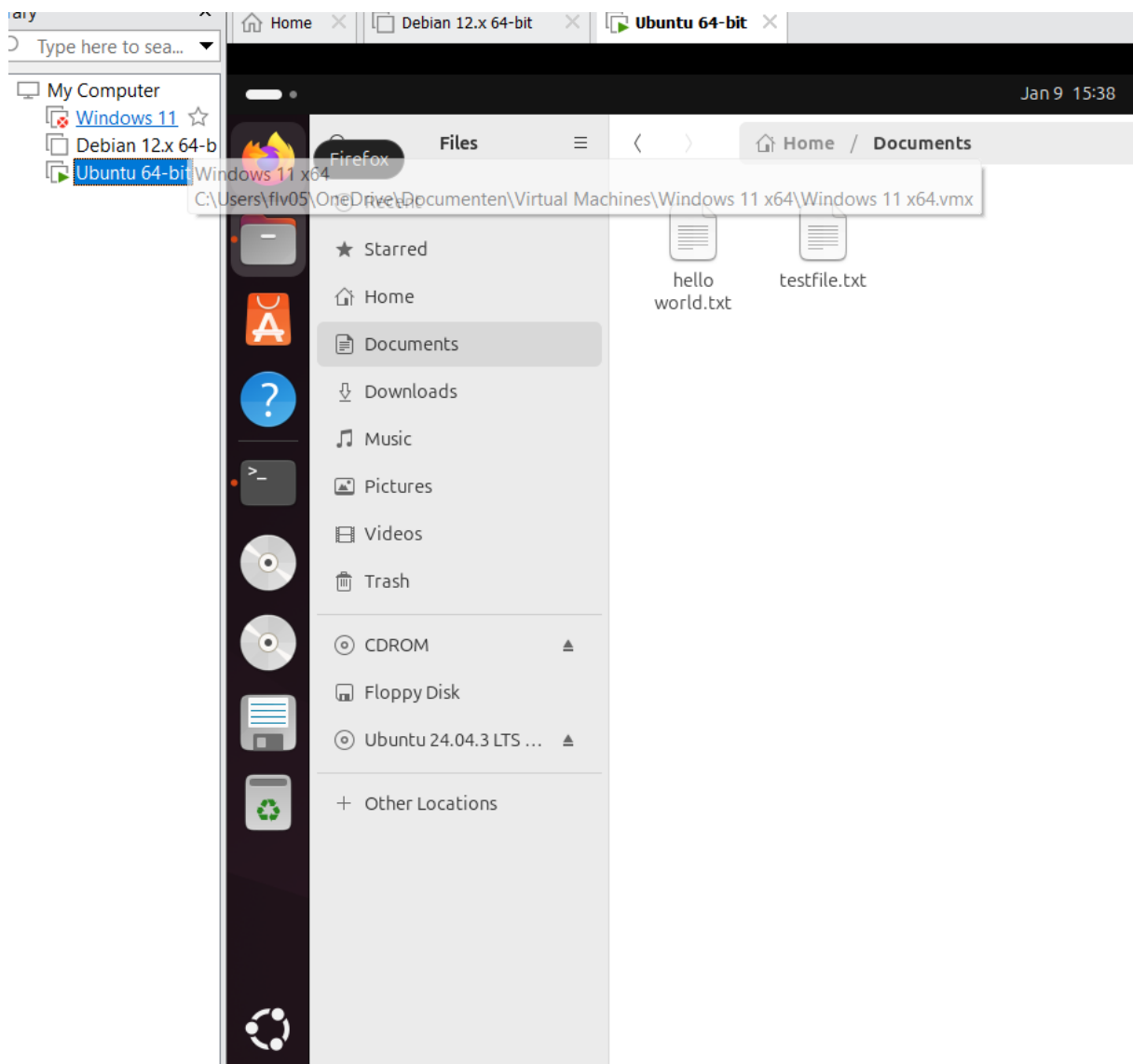
```
C:\Windows\System32>  
C:\Windows\System32>
```

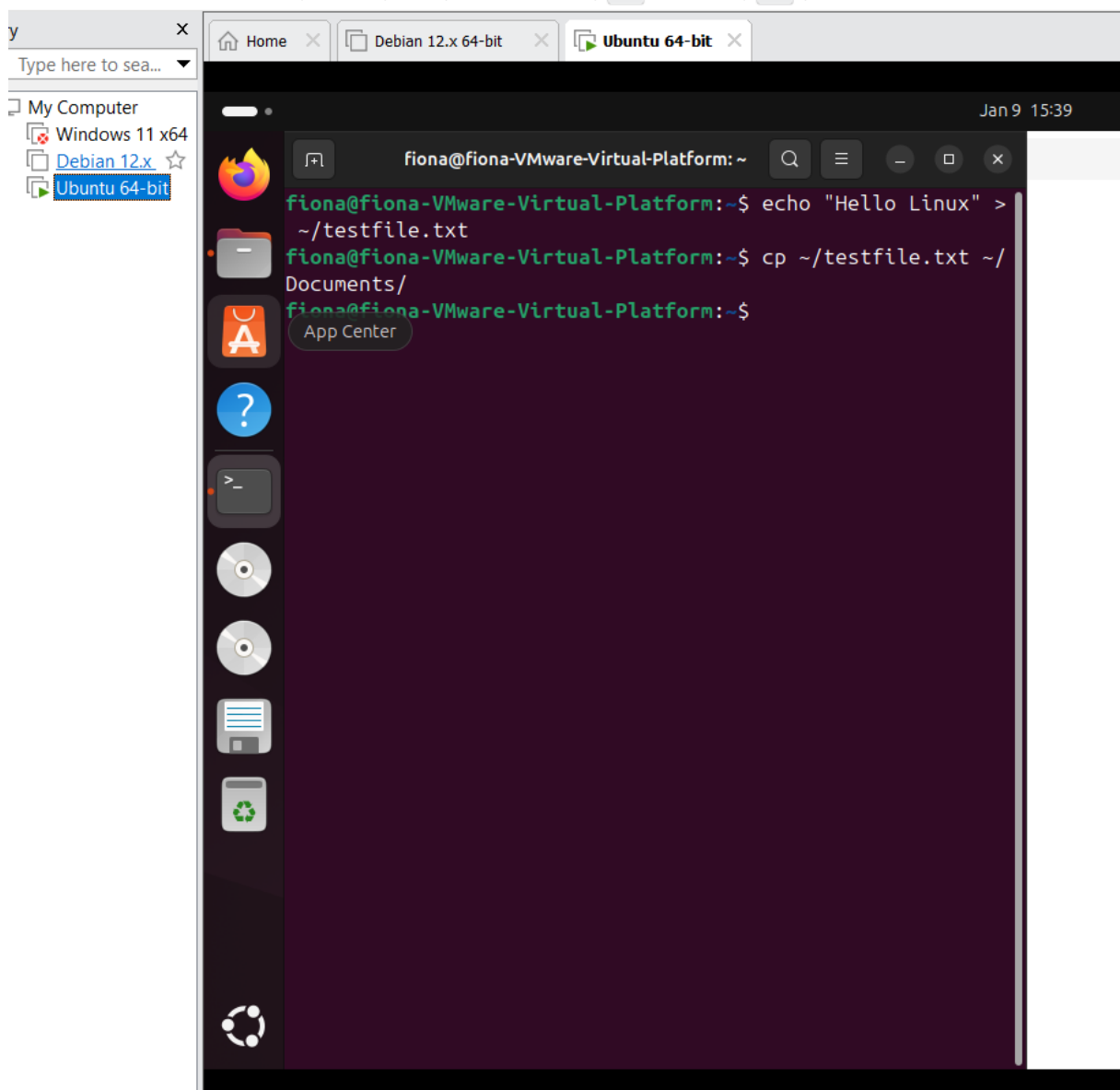
- WinSCP
- Notepad++
- 7zip

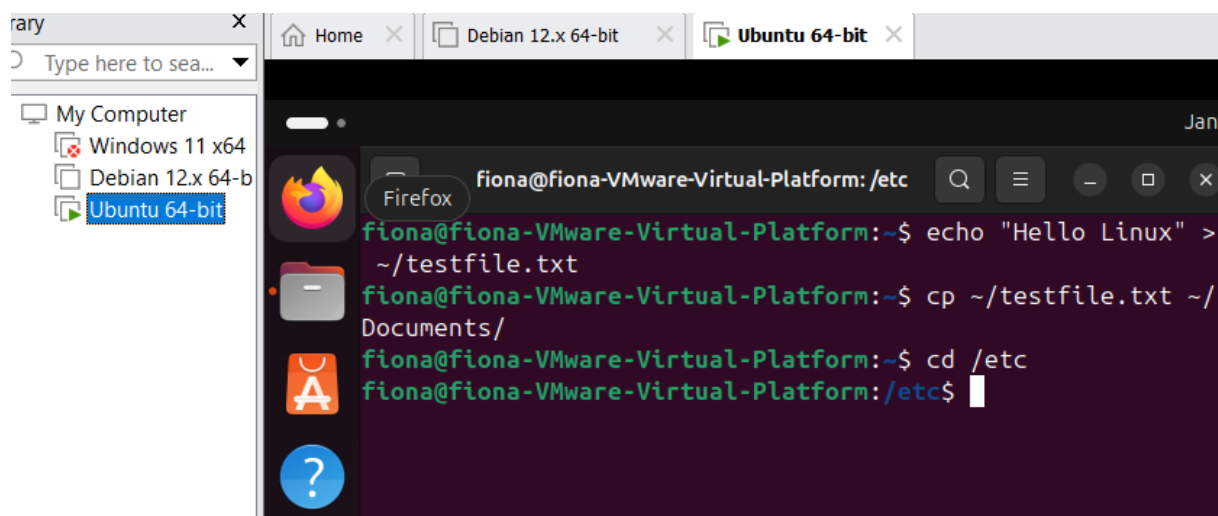
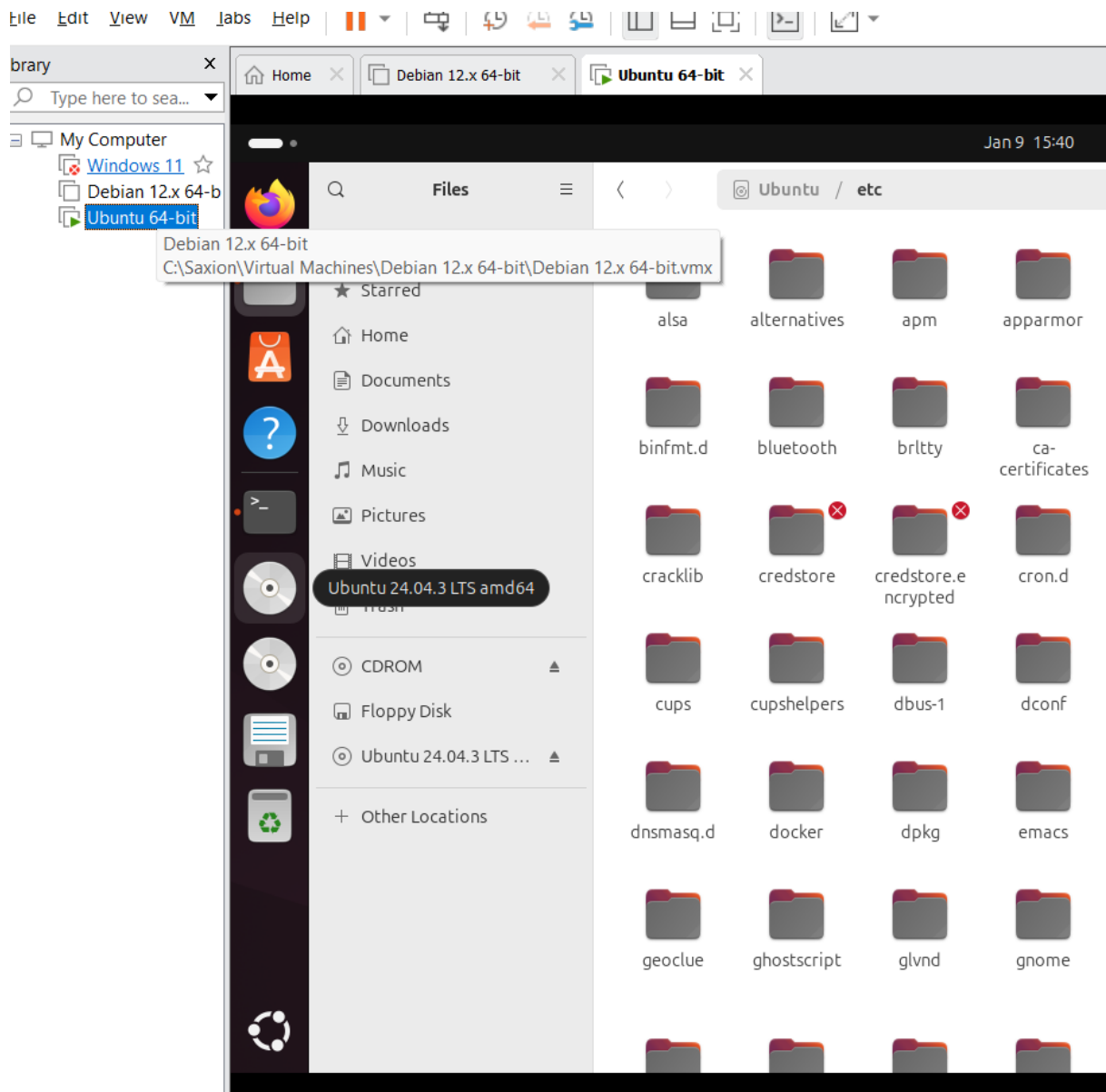
Assignment 5.4: Working with Linux

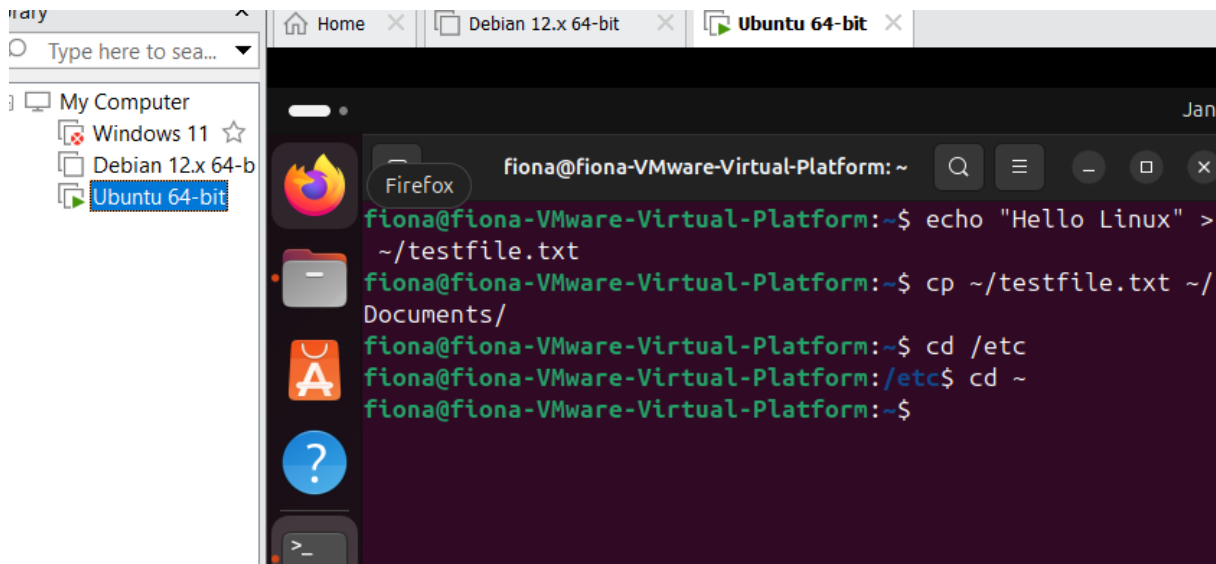
Relevant screenshots + motivation











Name one significant difference in Linux's file structure when comparing it to Windows.

Linux heeft een boomstructuur en gebruikt geen c:/

What is the /etc directory usually used for?

Heeft met configuratie bestanden te maken

Compress files o

Which command in the terminal would you use to compress a text file into a tar archive?

`tar -cvf archive.tar testfile.txt`

o With which command in the terminal would you be able to extract a tar file?

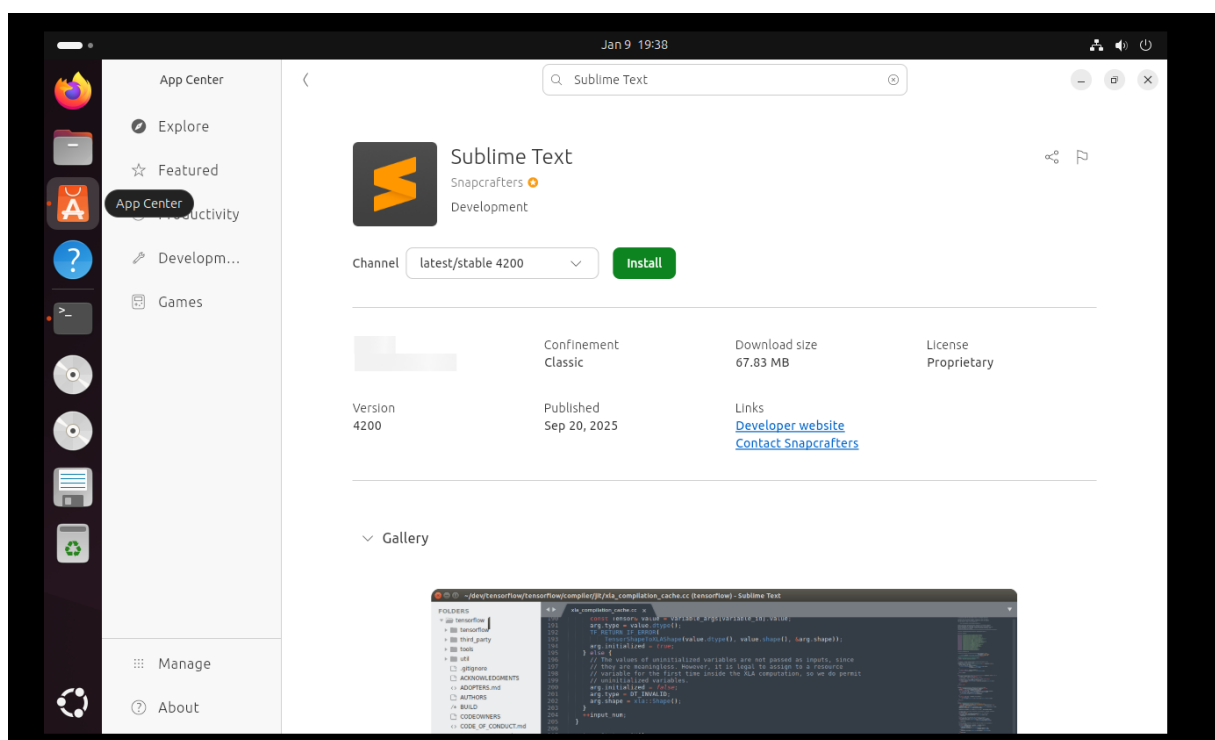
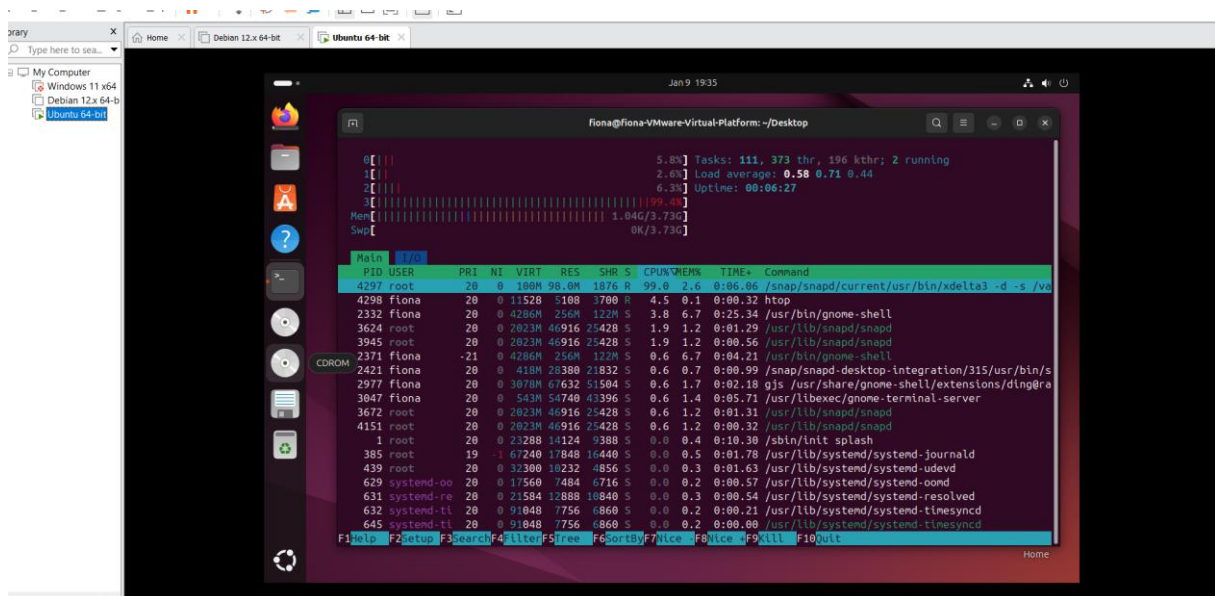
`tar -xvf archive.tar testfile.txt`

o Compress a text file in a tar archive and compress it with gzip.

`tar -cvzf archive.tar.gz testfile.txt`

Launch the htop application. Explain what this application shows

HTOP laat processen zien die draaien op de machine. Boven aan kan men de hoeveelheid CPU kracht zien, die er op het moment worden gebruikt. Daar onder staat “Mem”, dit geeft weer hoeveel RAM er wordt gebruikt. Daar onder vind je de “Swp” dit is swap memory. Swap memory wordt gebruikt als reserve opslag, als het werkgeheugen vol zit.



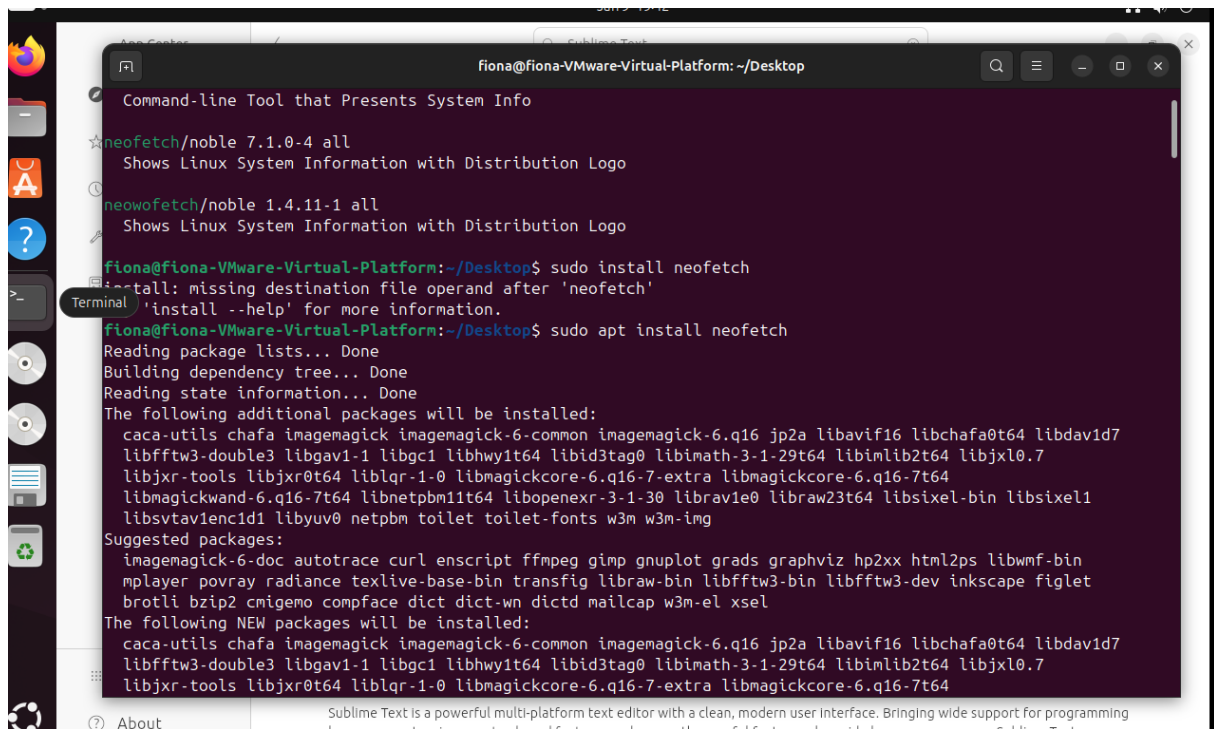
Using a terminal command, install the neofetch application. What does this application show when you launch it?

Neofetch laat system specificities zien en informatie over de desktop enviroment.

```
fiona@fiona-VMware-Virtual-Platform: ~/Desktop
fiona@fiona-VMware-Virtual-Platform:~/Desktop$ neofetch
fiona@fiona-VMware-Virtual-Platform
-----
OS: Ubuntu 24.04.3 LTS x86_64
Host: VMware Virtual Platform None
Kernel: 6.14.0-33-generic
Uptime: 13 mins
Packages: 1569 (dpkg), 12 (snap)
Shell: bash 5.2.21
Resolution: 1280x800
DE: GNOME 46.0
WM: Mutter
WM Theme: Adwaita
Theme: Yaru [GTK2/3]
Icons: Yaru [GTK2/3]
Terminal: gnome-terminal
CPU: Intel i5-10210U (4) @ 2.112GHz
GPU: 00:0F.0 VMware SVGA II Adapter
Memory: 1412MiB / 3819MiB

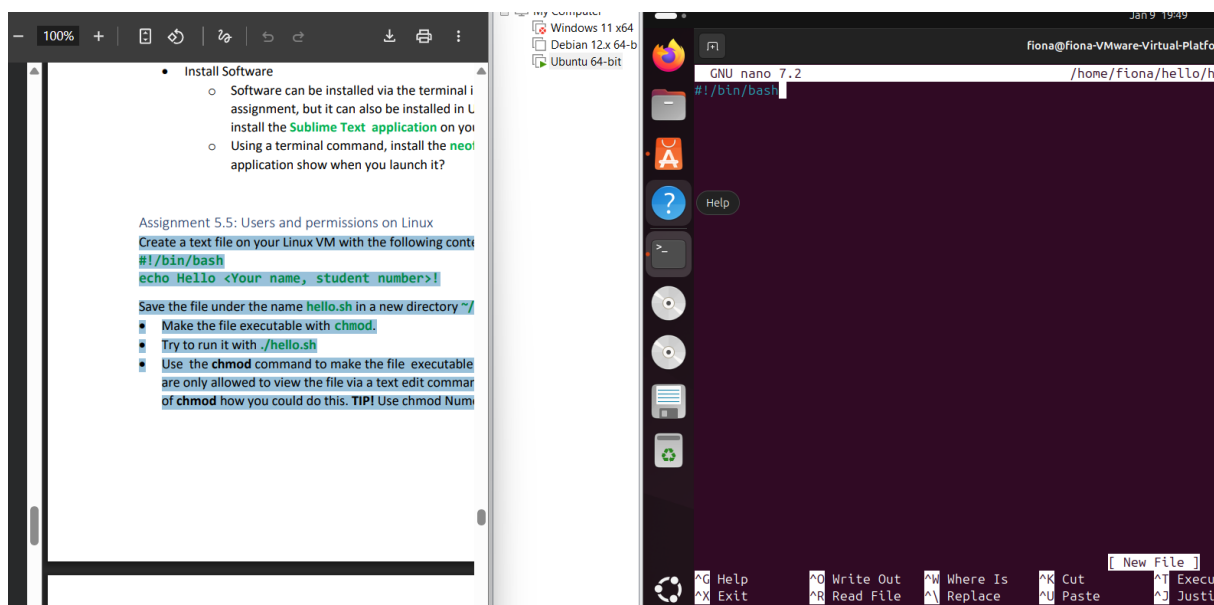
fiona@fiona-VMware-Virtual-Platform:~/Desktop$
```

```
fiona@fiona-VMware-Virtual-Platform: ~/Desktop
fiona@fiona-VMware-Virtual-Platform:~/Desktop$ sudo apt search neofetch
Sorting... Done
Full Text Search... Done
hyfetch/noble 1.4.11-1 all
  Command-line Tool that Presents System Info
neofetch/noble 7.1.0-4 all
  Shows Linux System Information with Distribution Logo
neowofetch/noble 1.4.11-1 all
  Shows Linux System Information with Distribution Logo
fiona@fiona-VMware-Virtual-Platform:~/Desktop$
```



```
fiona@fiona-VMware-Virtual-Platform: ~/Desktop
Command-line Tool that Presents System Info
☆neofetch/noble 7.1.0-4 all
Shows Linux System Information with Distribution Logo
neowofetch/noble 1.4.11-1 all
Shows Linux System Information with Distribution Logo
fiona@fiona-VMware-Virtual-Platform:~/Desktop$ sudo install neofetch
--tail: missing destination file operand after 'neofetch'
Terminal 'install --help' for more information.
fiona@fiona-VMware-Virtual-Platform:~/Desktop$ sudo apt install neofetch
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
caca-utils chafa imagemagick imagemagick-6-common imagemagick-6.q16 jp2a libavif16 libchafa0t64 libdav1d7
libbftw3-double3 libgav1-1 libgcl libhwy1t64 libid3tag0 libimath-3-1-29t64 libimlib2t64 libjxl0.7
libjxr-tools libjxr0t64 liblqr-1-0 libmagickcore-6.q16-7-extra libmagickcore-6.q16-7t64
libmagickwand-6.q16-7t64 libnetpbm11t64 libopenexr-3-1-30 libraw1e0 libraw23t64 libsixel-bin libsixel1
libsvtav1enc1d1 libyuv0 netpbm toilet toilet-fonts w3m w3m-img
Suggested packages:
imagemagick-6-doc autotrace curl enscript ffmpeg gimp gnuplot grads graphviz hp2xx html2ps libwmf-bin
mplayer povray radiance texlive-base-bin transfig libraw-bin libbftw3-bin libbftw3-dev inkscape figlet
brotli bzip2 cmigemo compface dict dict-wn dictd mailcap w3m-el xsel
The following NEW packages will be installed:
caca-utils chafa imagemagick imagemagick-6-common imagemagick-6.q16 jp2a libavif16 libchafa0t64 libdav1d7
libbftw3-double3 libgav1-1 libgcl libhwy1t64 libid3tag0 libimath-3-1-29t64 libimlib2t64 libjxl0.7
libjxr-tools libjxr0t64 liblqr-1-0 libmagickcore-6.q16-7-extra libmagickcore-6.q16-7t64
```

Assignment 5.5: Users and permissions on Linux



100% +

- Install Software
 - Software can be installed via the terminal i assignment, but it can also be installed in L install the **Sublime Text application** on you
 - Using a terminal command, install the **neo** application show when you launch it?

Assignment 5.5: Users and permissions on Linux

Create a text file on your Linux VM with the following cont

```
#!/bin/bash
echo Hello <Your name, student number>!
```

Save the file under the name **hello.sh** in a new directory ~/

- Make the file executable with **chmod**.
- Try to run it with **./hello.sh**
- Use the **chmod** command to make the file executable are only allowed to view the file via a text edit commar of **chmod** how you could do this. **TIP!** Use **chmod Num**

GNU nano 7.2

```
#!/bin/bash
```

Help

Ctrl+H Help Ctrl+O Write Out Ctrl+W Where Is Ctrl+K Cut Ctrl+U Paste Ctrl+J Justif

```
fiona@fiona-VMware-Virtual-Platform: ~/Desktop
fiona@fiona-VMware-Virtual-Platform:~/Desktop$ neofetch
      .-/+oosssso+/-.
      `:+ssssssssssss++:`
      ++ssssssssssssssyyssss++
      /sssssssssssssssdMMNNyssssso.
      /ssssssssssshdnNNNNyNNMMNhsssss+/
      +ssssssssshmydNNNNNNMddddy+ssssss++
      /ssssssssshNNMMNhhhyyyhNNMMNhsssss+/
      .ssssssssshNNMMNhsssssssshNNMMd+ssssss+.
      +ssssshhhyNNMMNysssssssssyNNMMNyssssss++
      oosyNNMMNyMMNhsssssssssssshNNMhsssssso
      oosyNNMMNyMMNhsssssssssssshNNMhsssssso
      +ssssshhhyNNMMNysssssssssyNNMMNyssssss++
      .ssssssssshNNMMNhsssssssshNNMMd+ssssss+.
      /ssssssssshNNMMNhhhyyyhNNMMNhsssss+/
      +ssssssssshmydNNNNNNMddddy+ssssss++
      /ssssssssshdnNNNNyNNMMNhsssss+/
      .sssssssssssssssdMMNNyssssso.
      ++ssssssssssssssyyssss++
      `:+ssssssssssss++:`
      .-/+oosssso+/-.

      OS: Ubuntu 24.04.3 LTS x86_64
      Host: VMware Virtual Platform None
      Kernel: 6.14.0-33-generic
      Uptime: 13 mins
      Packages: 1569 (dpkg), 12 (snap)
      Shell: bash 5.2.21
      Resolution: 1280x800
      DE: GNOME 46.0
      WM: Mutter
      WM Theme: Adwaita
      Theme: Yaru [GTK2/3]
      Icons: Yaru [GTK2/3]
      Terminal: gnome-terminal
      CPU: Intel i5-10210U (4) @ 2.112GHz
      GPU: 00:0f:0 VMware SVGA II Adapter
      Memory: 1412MiB / 3819MiB

fiona@fiona-VMware-Virtual-Platform:~/Desktop$ mkdir ~/hello
fiona@fiona-VMware-Virtual-Platform:~/Desktop$ nano ~/hello/hello.sh
fiona@fiona-VMware-Virtual-Platform:~/Desktop$ chmod +x ~/hello/hello.sh
fiona@fiona-VMware-Virtual-Platform:~/Desktop$
```

```
fiona@fiona-VMware-Virtual-Platform: ~/hello
      .-/+oosssso+/-.
      `:+ssssssssssss++:`
      ++ssssssssssssssyyssss++
      /sssssssssssssssdMMNNyssssso.
      /ssssssssssshdnNNNNyNNMMNhsssss+/
      +ssssssssshmydNNNNNNMddddy+ssssss++
      /ssssssssshNNMMNhhhyyyhNNMMNhsssss+/
      .ssssssssshNNMMNhsssssssshNNMMd+ssssss+.
      +ssssshhhyNNMMNysssssssssyNNMMNyssssss++
      oosyNNMMNyMMNhsssssssssssshNNMhsssssso
      oosyNNMMNyMMNhsssssssssssshNNMhsssssso
      +ssssshhhyNNMMNysssssssssyNNMMNyssssss++
      .ssssssssshNNMMNhsssssssshNNMMd+ssssss+.
      /ssssssssshNNMMNhhhyyyhNNMMNhsssss+/
      +ssssssssshmydNNNNNNMddddy+ssssss++
      /ssssssssshdnNNNNyNNMMNhsssss+/
      .sssssssssssssssdMMNNyssssso.
      ++ssssssssssssssyyssss++
      `:+ssssssssssss++:`
      .-/+oosssso+/-.

      OS: Ubuntu 24.04.3 LTS x86_64
      Host: VMware Virtual Platform None
      Kernel: 6.14.0-33-generic
      Uptime: 13 mins
      Packages: 1569 (dpkg), 12 (snap)
      Shell: bash 5.2.21
      Resolution: 1280x800
      DE: GNOME 46.0
      WM: Mutter
      WM Theme: Adwaita
      Theme: Yaru [GTK2/3]
      Icons: Yaru [GTK2/3]
      Terminal: gnome-terminal
      CPU: Intel i5-10210U (4) @ 2.112GHz
      GPU: 00:0f:0 VMware SVGA II Adapter
      Memory: 1412MiB / 3819MiB

fiona@fiona-VMware-Virtual-Platform:~/Desktop$ mkdir ~/hello
fiona@fiona-VMware-Virtual-Platform:~/Desktop$ nano ~/hello/hello.sh
fiona@fiona-VMware-Virtual-Platform:~/Desktop$ chmod +x ~/hello/hello.sh
fiona@fiona-VMware-Virtual-Platform:~/Desktop$ cd hello
bash: cd: hello: No such file or directory
fiona@fiona-VMware-Virtual-Platform:~/Desktop$ cd ../hello
fiona@fiona-VMware-Virtual-Platform:~/hello$ ls
hello.sh
fiona@fiona-VMware-Virtual-Platform:~/hello$ ./hello.sh
Hello fiona 583809
fiona@fiona-VMware-Virtual-Platform:~/hello$
```

Ik gebruik mkdir om een map te maken

Nano gebruik ik om de tekst te kunnen editen

Chmod ~+ is de rechten die ik geef zorgt ervoor dat iedereen rechten heeft

Assignment 5.6: View the contents of files

cat filename.txt: geeft alle data uit een bestand weer in de terminal

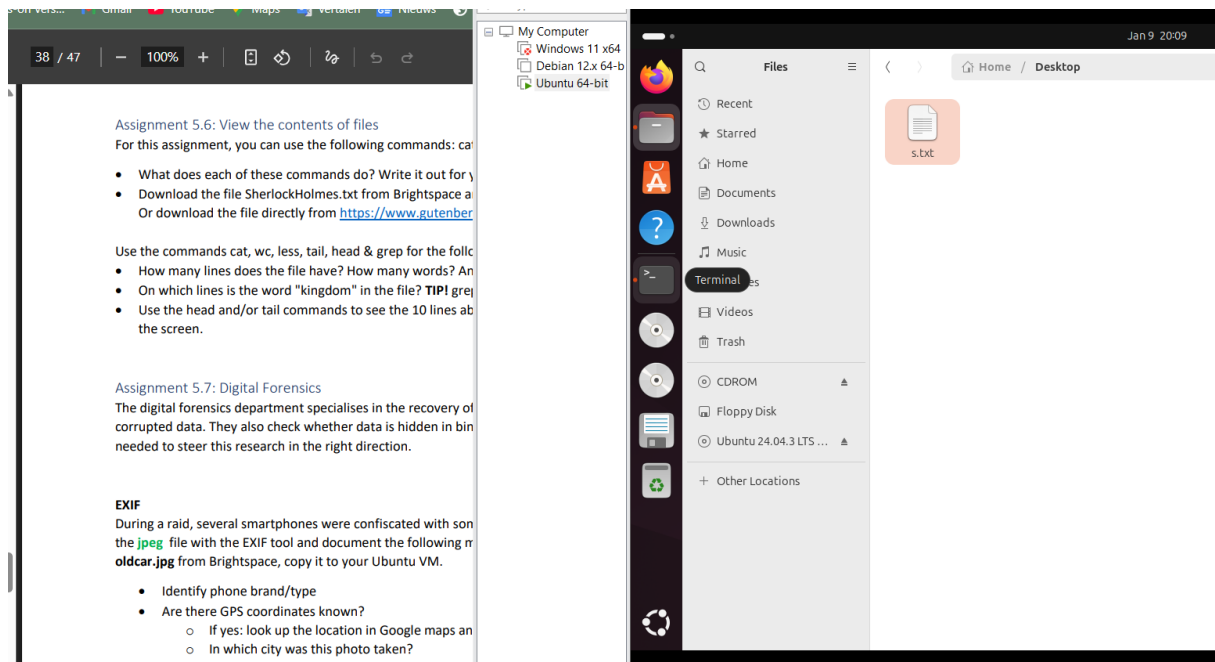
wc -l file.txt laat het aantal lijnen in een bestaan zien

less file.txt laat de content van file.txt zien in 1 scherm, zelfde als “more” maar met meer mogelijkheden.

Tail file.txt laat de laatste 10 lijnen van file.txt zien in terminal

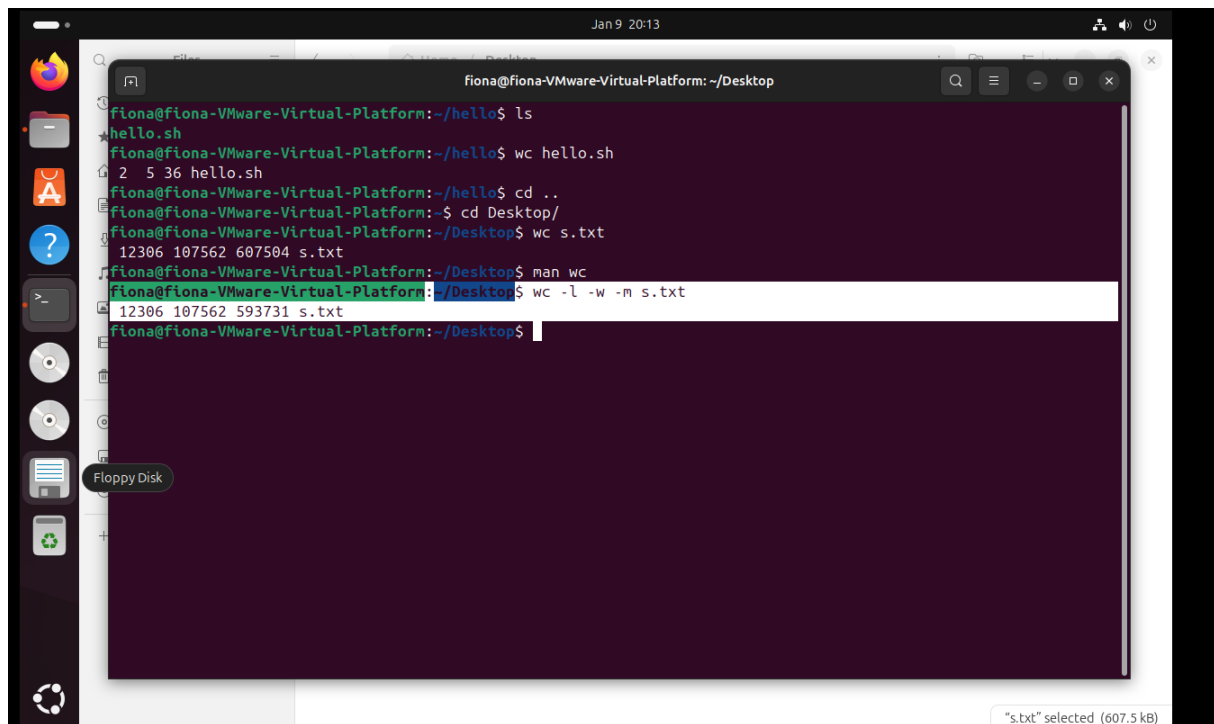
Head file.txt laat de eerste 10 lijnen zien van file.txt

grep is handig om data naar toe te pipen e.g: “ls -la | grep hallo” laat alle bestanden en dirs zien met “hallo” in de naam.



Use the commands cat, wc, less, tail, head & grep for the following assignments:

- How many lines does the file have? How many words? And how many characters?



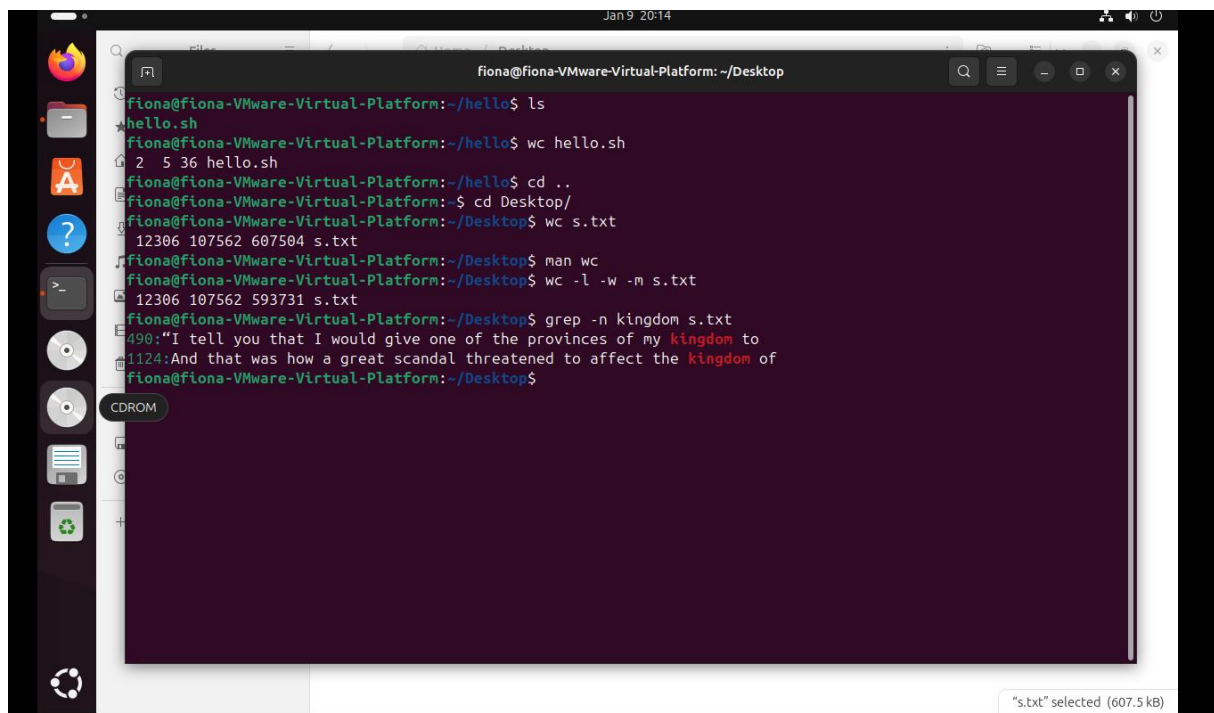
A terminal window titled 'fiona@fiona-VMware-Virtual-Platform: ~/Desktop' showing the following commands and output:

```
fiona@fiona-VMware-Virtual-Platform:~/hello$ ls
hello.sh
fiona@fiona-VMware-Virtual-Platform:~/hello$ wc hello.sh
2  5 36 hello.sh
fiona@fiona-VMware-Virtual-Platform:~/hello$ cd ..
fiona@fiona-VMware-Virtual-Platform:~$ cd Desktop/
fiona@fiona-VMware-Virtual-Platform:~/Desktop$ wc s.txt
12306 107562 607504 s.txt
fiona@fiona-VMware-Virtual-Platform:~/Desktop$ man wc
fiona@fiona-VMware-Virtual-Platform:~/Desktop$ wc -l -w -m s.txt
12306 107562 593731 s.txt
fiona@fiona-VMware-Virtual-Platform:~/Desktop$
```

The terminal window is part of a desktop environment with a sidebar on the left containing icons for applications like Firefox, Files, and a terminal. A 'Floppy Disk' icon is visible in the sidebar. The window title bar shows the date and time as 'Jan 9 20:13'.

Wc lines, word, characters

- On which lines is the word "kingdom" in the file? TIP! `grep -n`



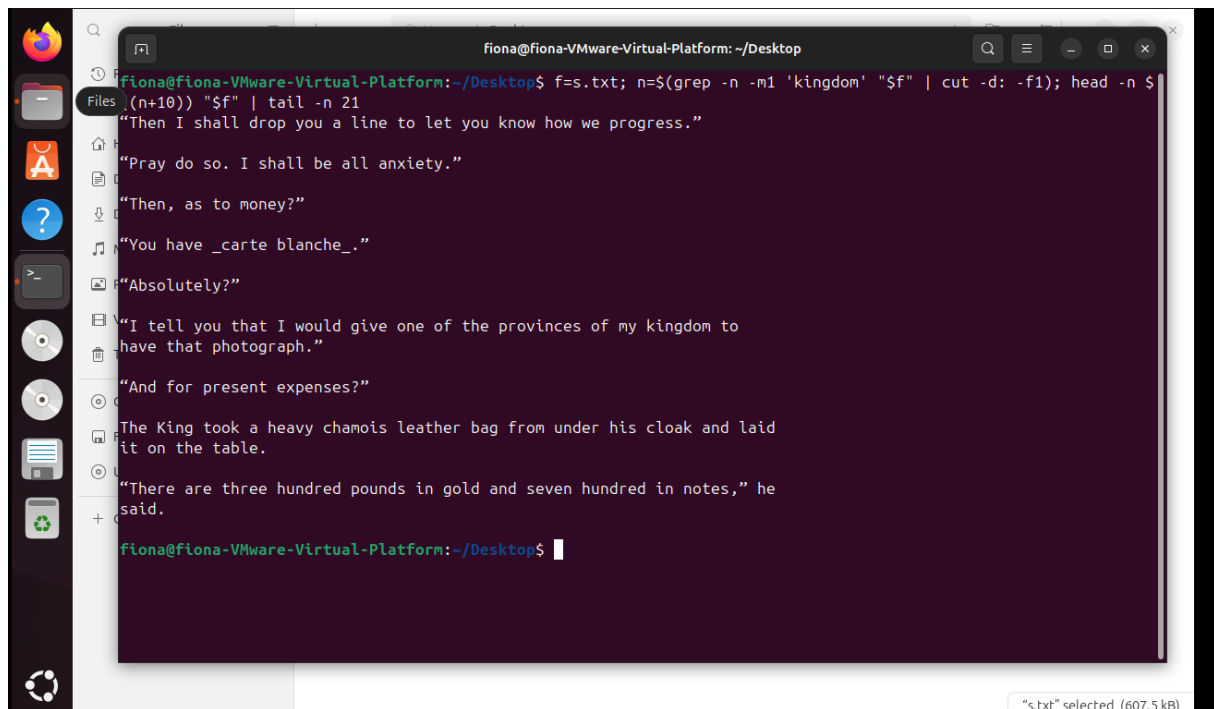
A terminal window titled 'fiona@fiona-VMware-Virtual-Platform: ~/Desktop' showing the following commands and output:

```
fiona@fiona-VMware-Virtual-Platform:~/hello$ ls
hello.sh
fiona@fiona-VMware-Virtual-Platform:~/hello$ wc hello.sh
2  5 36 hello.sh
fiona@fiona-VMware-Virtual-Platform:~/hello$ cd ..
fiona@fiona-VMware-Virtual-Platform:~$ cd Desktop/
fiona@fiona-VMware-Virtual-Platform:~/Desktop$ wc s.txt
12306 107562 607504 s.txt
fiona@fiona-VMware-Virtual-Platform:~/Desktop$ man wc
fiona@fiona-VMware-Virtual-Platform:~/Desktop$ wc -l -w -m s.txt
12306 107562 593731 s.txt
fiona@fiona-VMware-Virtual-Platform:~/Desktop$ grep -n kingdom s.txt
490:"I tell you that I would give one of the provinces of my kingdom to
1124:And that was how a great scandal threatened to affect the kingdom of
fiona@fiona-VMware-Virtual-Platform:~/Desktop$
```

The terminal window is part of a desktop environment with a sidebar on the left containing icons for applications like Firefox, Files, and a terminal. A 'CDROM' icon is visible in the sidebar. The window title bar shows the date and time as 'Jan 9 20:14'.

Op lijn 490 en lijn 1124

- Use the `head` and/or `tail` commands to see the 10 lines above and below the word "kingdom" on the screen.



```
fiona@fiona-VMware-Virtual-Platform: ~/Desktop
fiona@fiona-VMware-Virtual-Platform:~/Desktop$ f=s.txt; n=$(grep -n -m1 'kingdom' "$f" | cut -d: -f1); head -n $
(n+10)) "$f" | tail -n 21
"Then I shall drop you a line to let you know how we progress."
"Pray do so. I shall be all anxiety."
"Then, as to money?"
"You have _carte blanche_."
"Absolutely?"
"I tell you that I would give one of the provinces of my kingdom to
have that photograph."
"And for present expenses?"
The King took a heavy chamois leather bag from under his cloak and laid
it on the table.
"There are three hundred pounds in gold and seven hundred in notes," he
said.
fiona@fiona-VMware-Virtual-Platform:~/Desktop$
```

Vind de eerste lijnen die kingdom bevatten

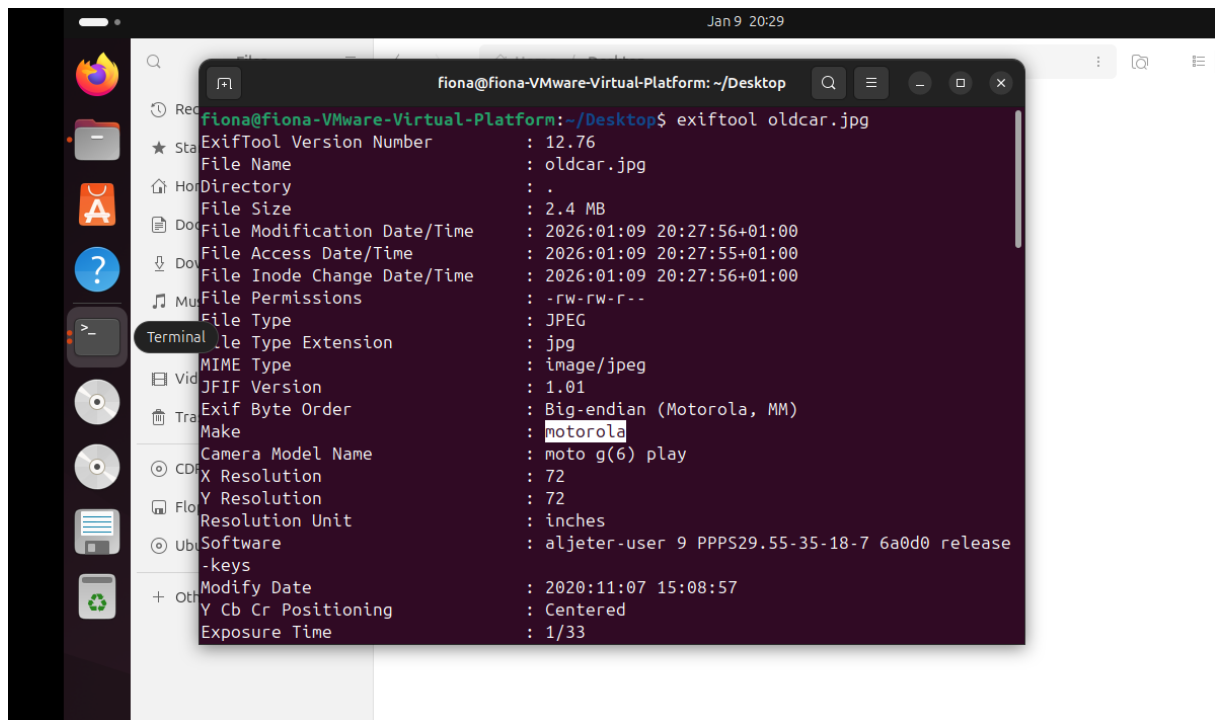
Print tot maximaal 10 lijnen onder die zin met head -n n+10

Houd de laatste 21 lijnen (10 voor + kingdom + 10 er na) met tail -n 21

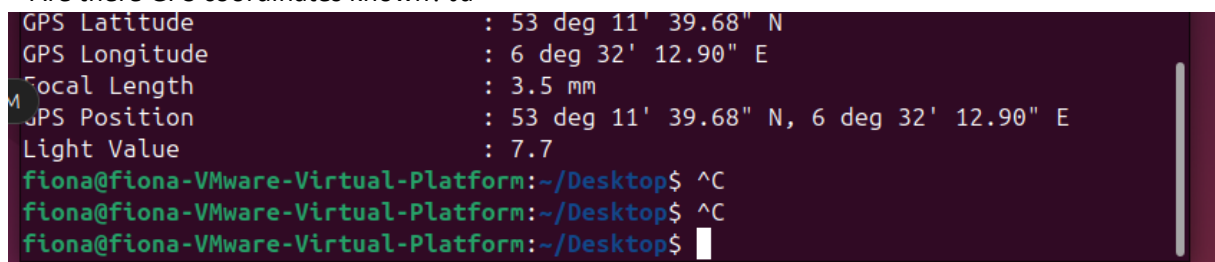
Assignment 5.7: Digital forensics

Relevant screenshots + motivation

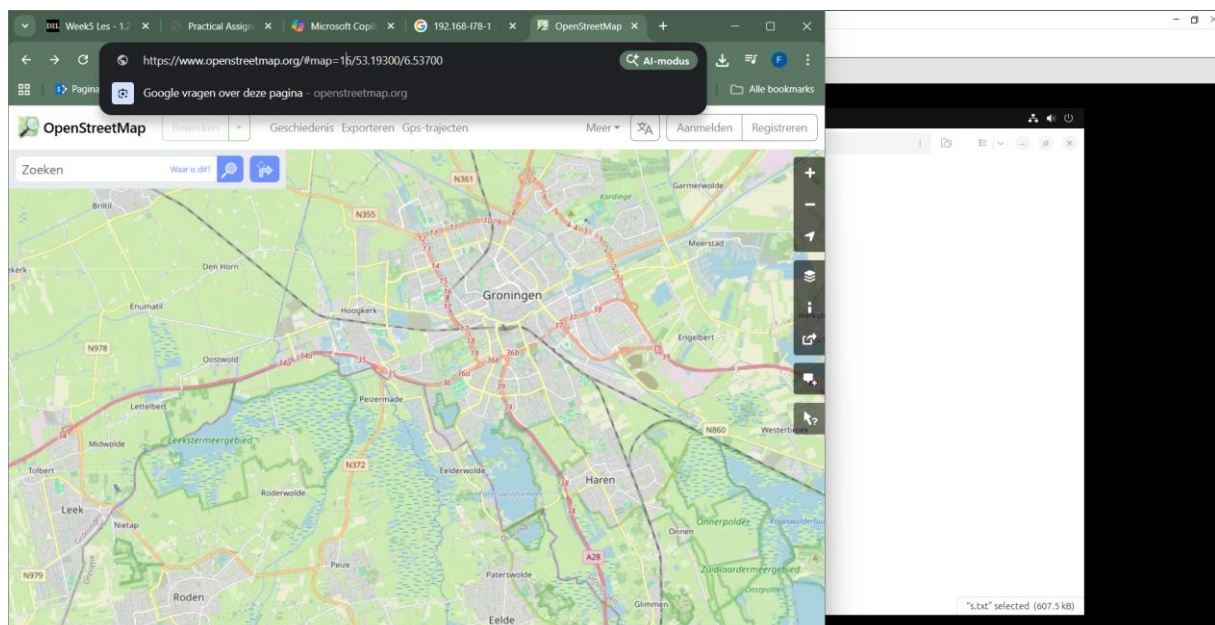
Identify phone brand/type



- Are there GPS coordinates known? Ja



If yes: look up the location in Google maps and Streetview



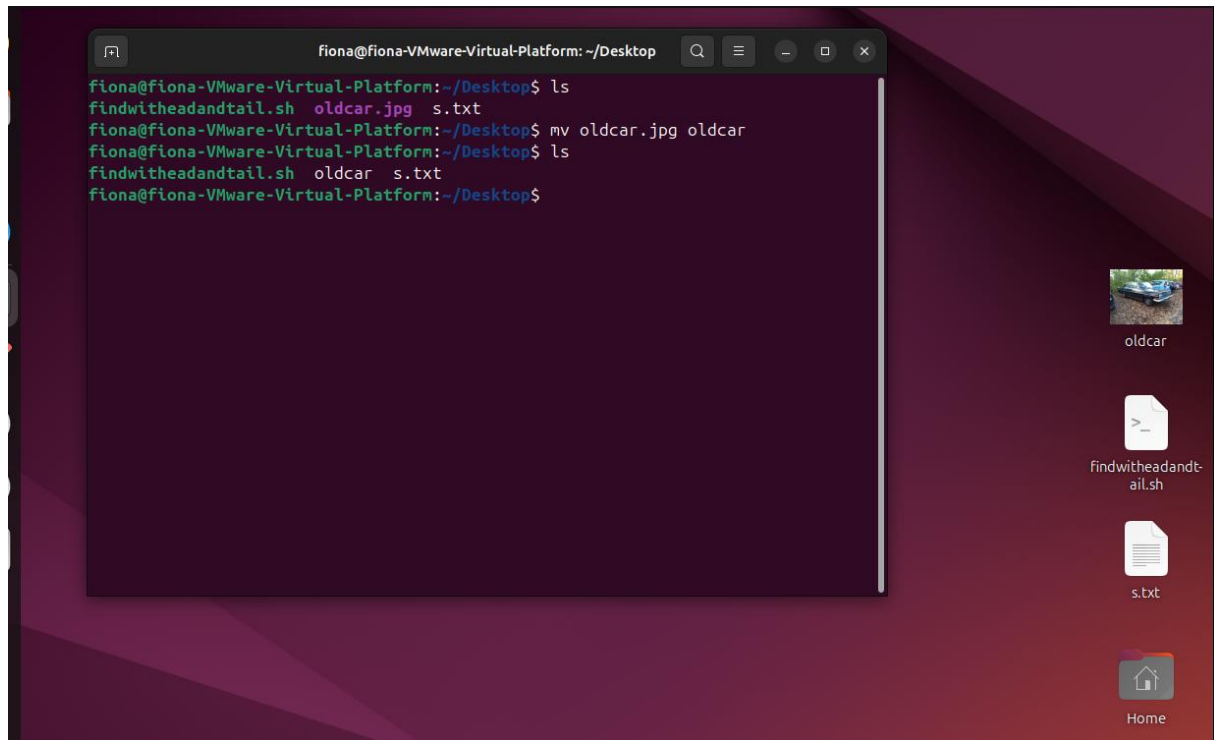
In which city was this photo taken?

In Groningen

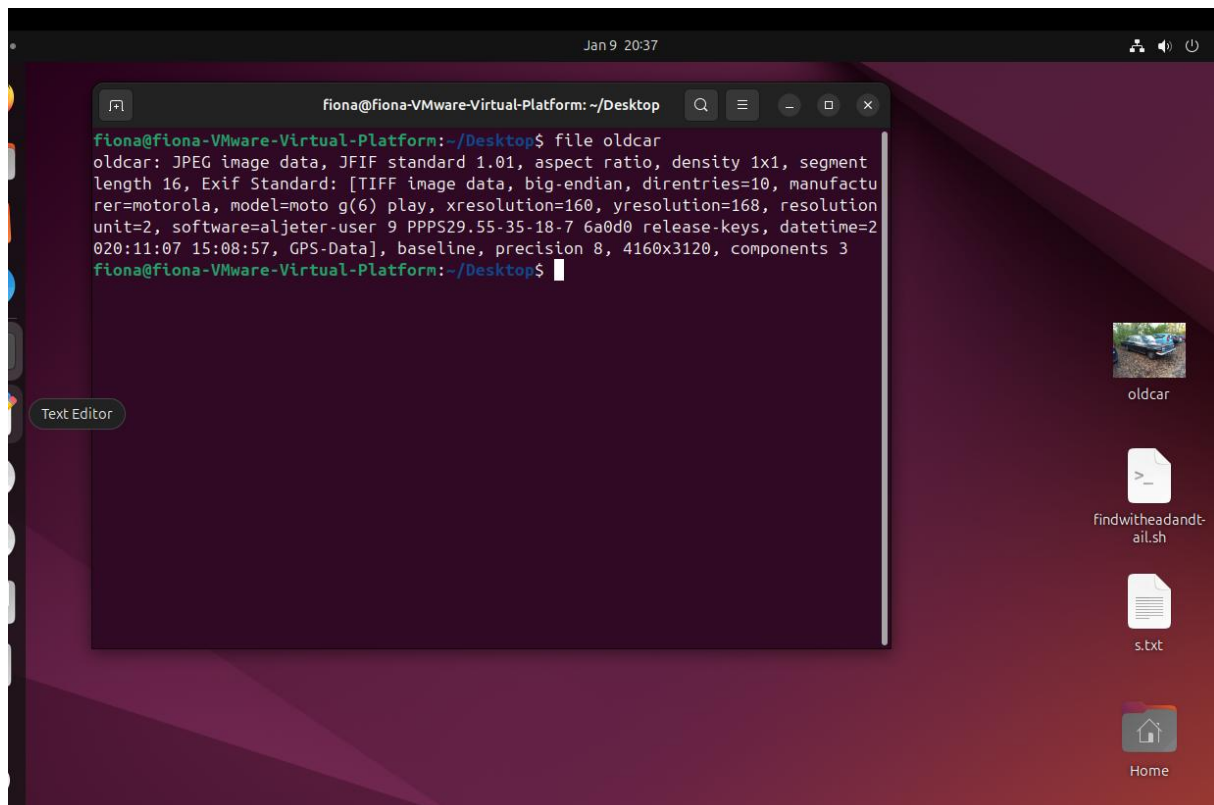
in Windows, the file extensions are important because they determine the file type. Windows only looks at the extension and then tries to open that file with the default application that is suitable for it.

While in the Ubuntu VM we could remove the extension of the oldcar.jpg file and it will still be opened as a jpg file by an image viewer.

- Rename the file to oldcar. (So you've removed the file extension)



- In the terminal, type the command file oldcar.

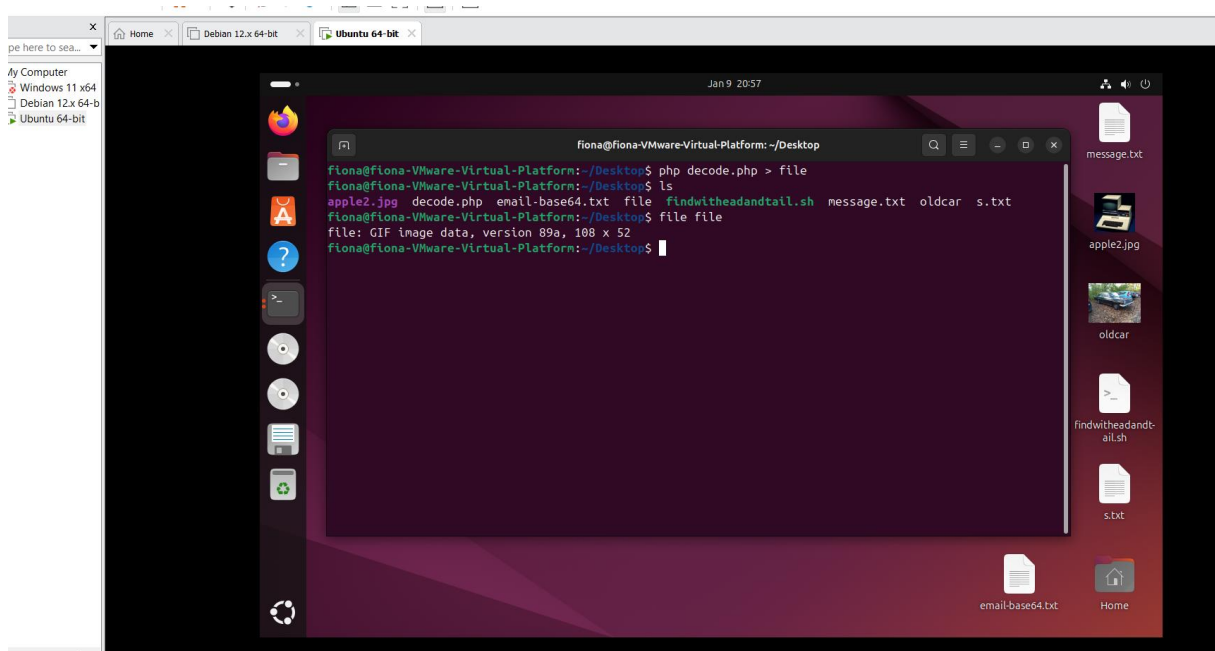


- Does Ubuntu still consider it to be a jpg file?

Ja door doordat hij info terug geeft met jpg image data.

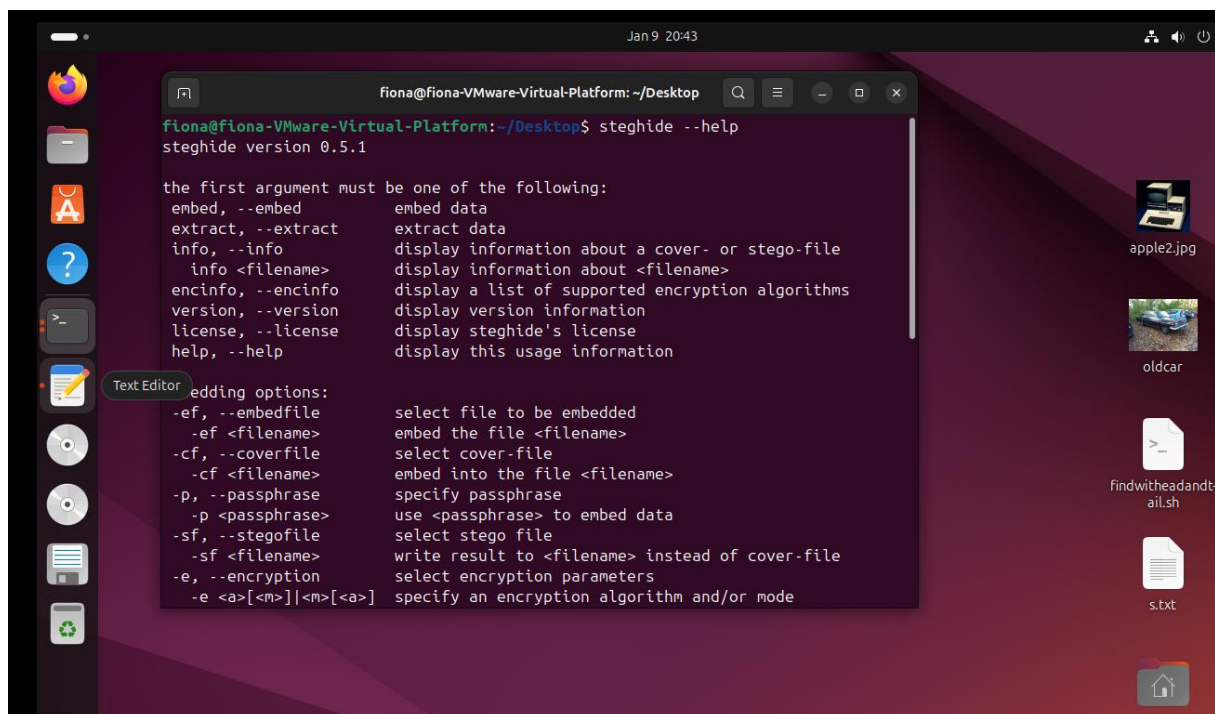
BASE64 Hexadecimal is also called BASE16, which uses the characters 0-9 and A-F to encode binary data. We have seen this in the previous exercises. Another widely used system for encoding binary data is BASE64. That's what we're going to work with in this assignment. A partial e-mail message has been intercepted containing binary data. Binary data sent via e-mail is often encrypted via BASE64. During the BASE64 encoding process, groups of 3 x 8-bit will be regrouped into groups of 4 x 6-bit and translated into characters A-Z, a-z, 0-9 and +/ . The = sign is also used as a filler sign.

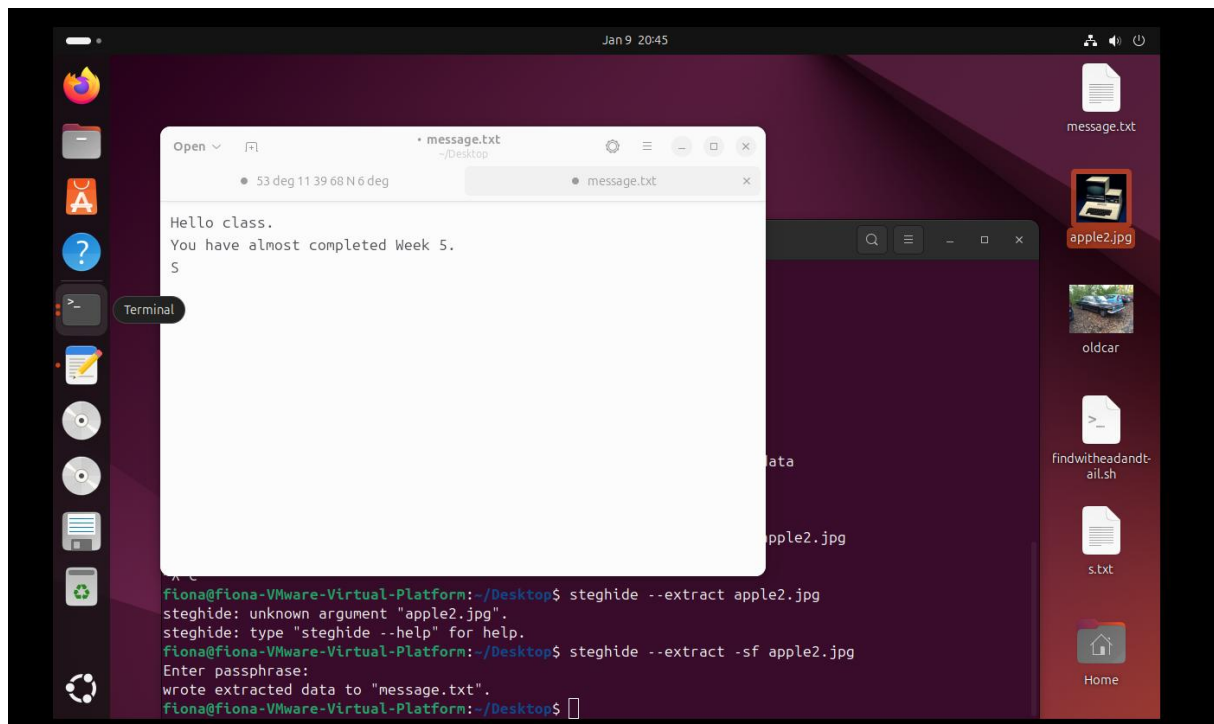
Download the following text file from Brightspace and copy it to your Ubuntu VM: email-base64.txt
You can also see the contents of this file below. A colleague of yours who has a lot of experience with BASE64 data thinks that this is a gif image file



De collega had gelijk.

Assignment 5.8: Steganography





Ik heb de tekst bestand eruit gehaald.

Assignment 5.9: Capture disk images

Make relevant screenshots + motivation:

- Proof that the Debian 13 server stored a back-up image of the Ubuntu 24.04 Desktop VM.
- Proof that you can restore the back-up image into an empty VM.

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